

ME400: PROJECT AND THESIS

Multiphysics Modeling of Hybrid Laminated Composite Conductors under a DC Field

presented by

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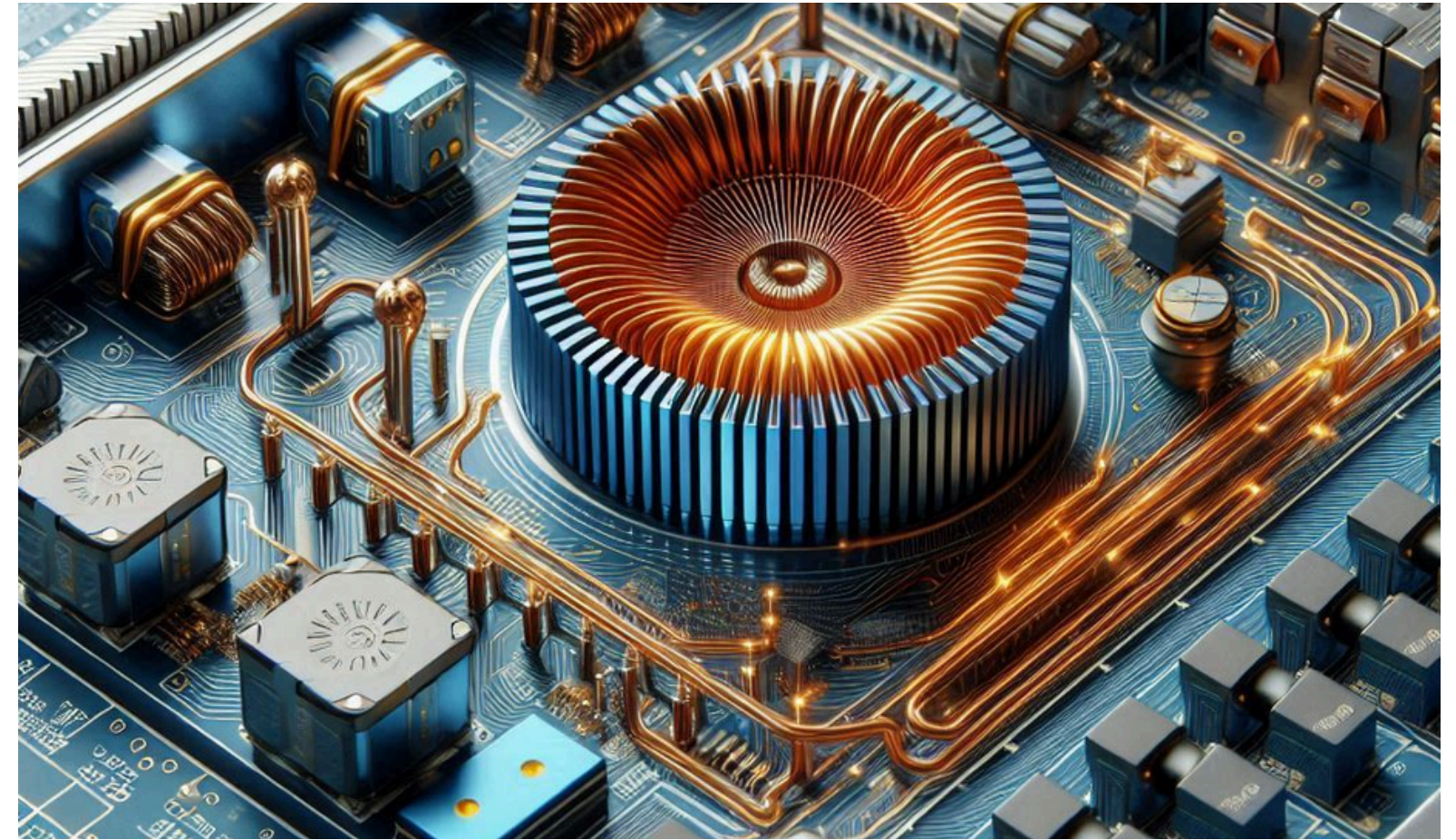
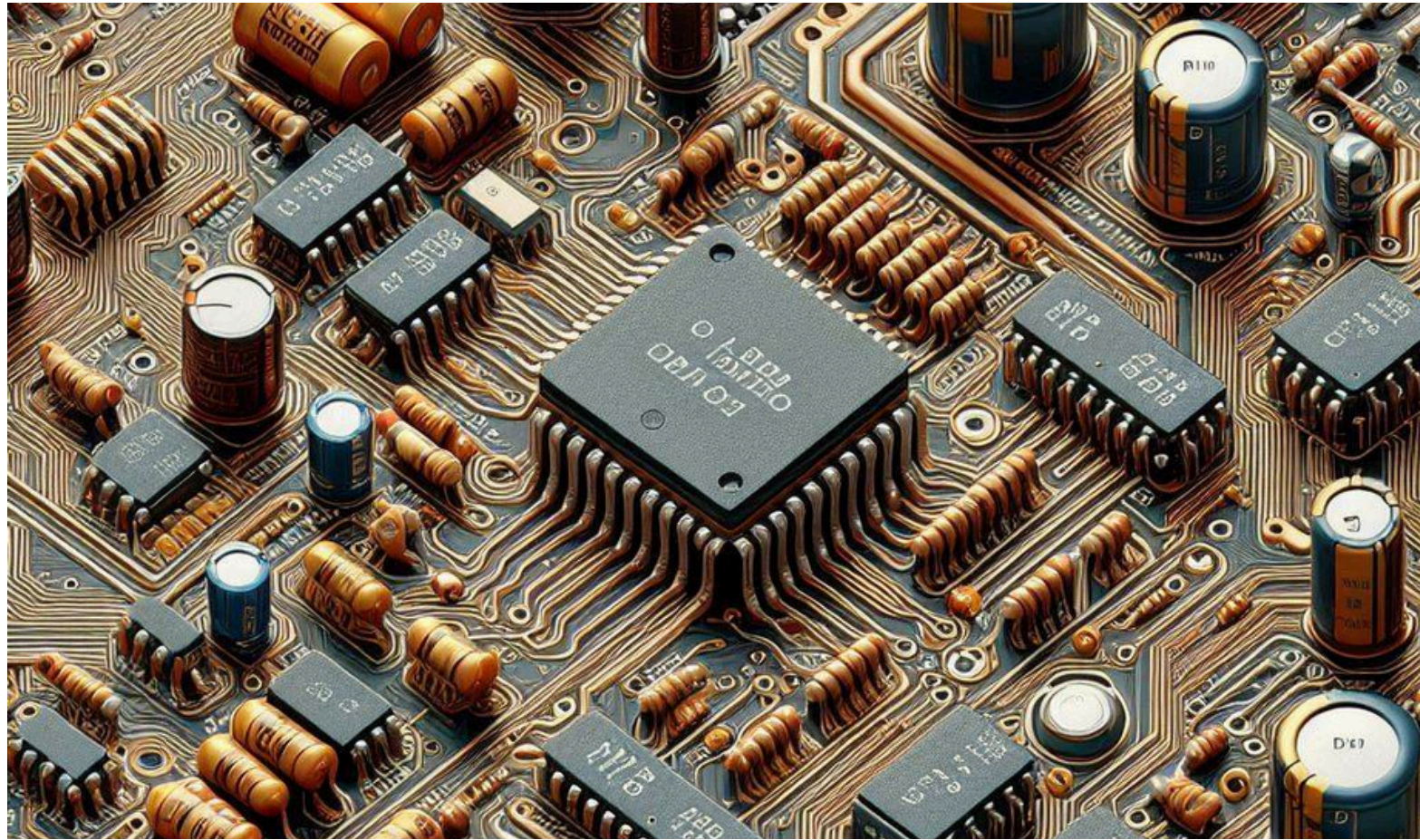


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Motivation



- **Advancing Material Technologies for Next-Generation Electronics**
 - Modern electronics, aerospace, and power systems **demand materials that are efficient, lightweight, and thermally stable.**
 - Conventional metal conductors face limitations in heat dissipation and electrical performance, necessitating advanced material solutions.

Motivation

- **Why Laminated Composites?**

- Hybrid laminated structures enhance electrical, thermal, and mechanical properties
- **Weight reduction & tailorable material composition** for specific performance requirements

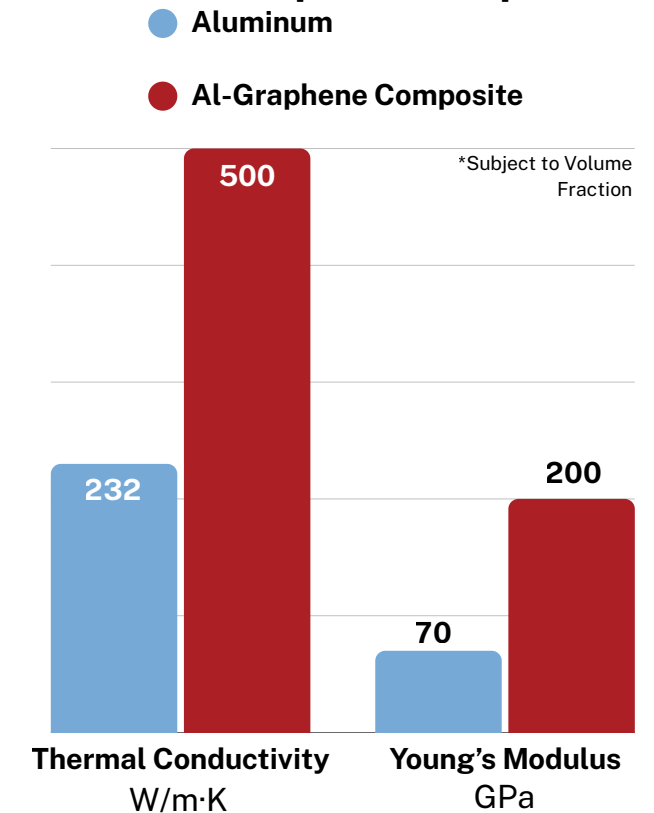
- **Advantages of Displacement Function Modeling**

- **Efficient Computation** – Reduces variables for faster, cost-effective solutions.
- **Accurate Stress Prediction** – Ensures smooth stress distribution in composites.

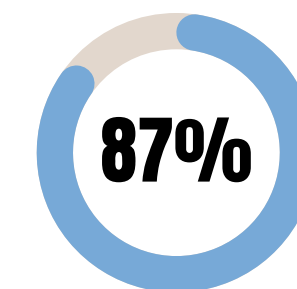
- **Limitations of Finite Element Method**

- **High computational cost** for fine meshes
- Requires **extensive meshing for sharp gradients** in electrical and thermal fields

Pure Metal vs Composite Properties



Displacement Function Advantage:



Reduction in computational effort
by using Displacement Function
Models compared to FEA.

S. R. Ahmed et al., 2005.

Literature Review



1960-2000

Development of Theoretical and Numerical Models

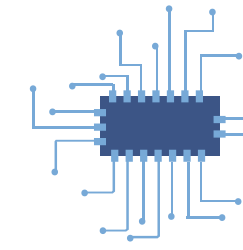
- Zienkiewicz et al. 1960 – Stress function formulation for isotropic materials.
- Dundurs et al. 1970 – Stress singularities in mixed boundary problems.
- Ahmed, Akanda et al. 2000 – Finite difference modeling of mixed-boundary-value problems.



2000-2020

Application to Composite Structures

- Ahmed et al. 2005 – Displacement potential function applied to anisotropic materials.
- Nath, Ahmed et al. 2015 – Extended method for fiber-reinforced composites.
- Bordoloi et al. 2019 – Analysis of Stress Through Thickness of Hybrid Laminates.



2021-Present

Emerging Research on Conducting Composites

- Ahmed, Sarkar 2021 – Thermo-mechanical analysis of fiber-reinforced composites.
- Zheng et. al. 2022 – Fabrication of Cu/Al/Cu Laminated Composites Reinforced with Graphene.
- Zahid et. al. 2023 – Conducting Polyaniline/Graphene Nanocomposites – Electromagnetic interference shielding applications.

Literature Review

Gap in Research

- **Complex Multiphysics Modeling:** Existing studies often employ intricate multiphysics models to analyze the behavior of laminated composites, leading to computational complexity and potential numerical instability.
- **Coupled Field Variables:** Traditional approaches typically involve solving coupled field variables simultaneously, which can complicate the modeling process and increase computational demands.
- **Layer-wise Theories:** Some methodologies utilize layerwise theories that require separate governing equations for each layer, adding to the complexity and computational burden.

Our Contribution

“This study is the first in

- modelling the electro-thermo-mechanical response of laminated composites
- **using a single governing variable per field**—voltage for electrical, temperature for thermal, and displacement for mechanical

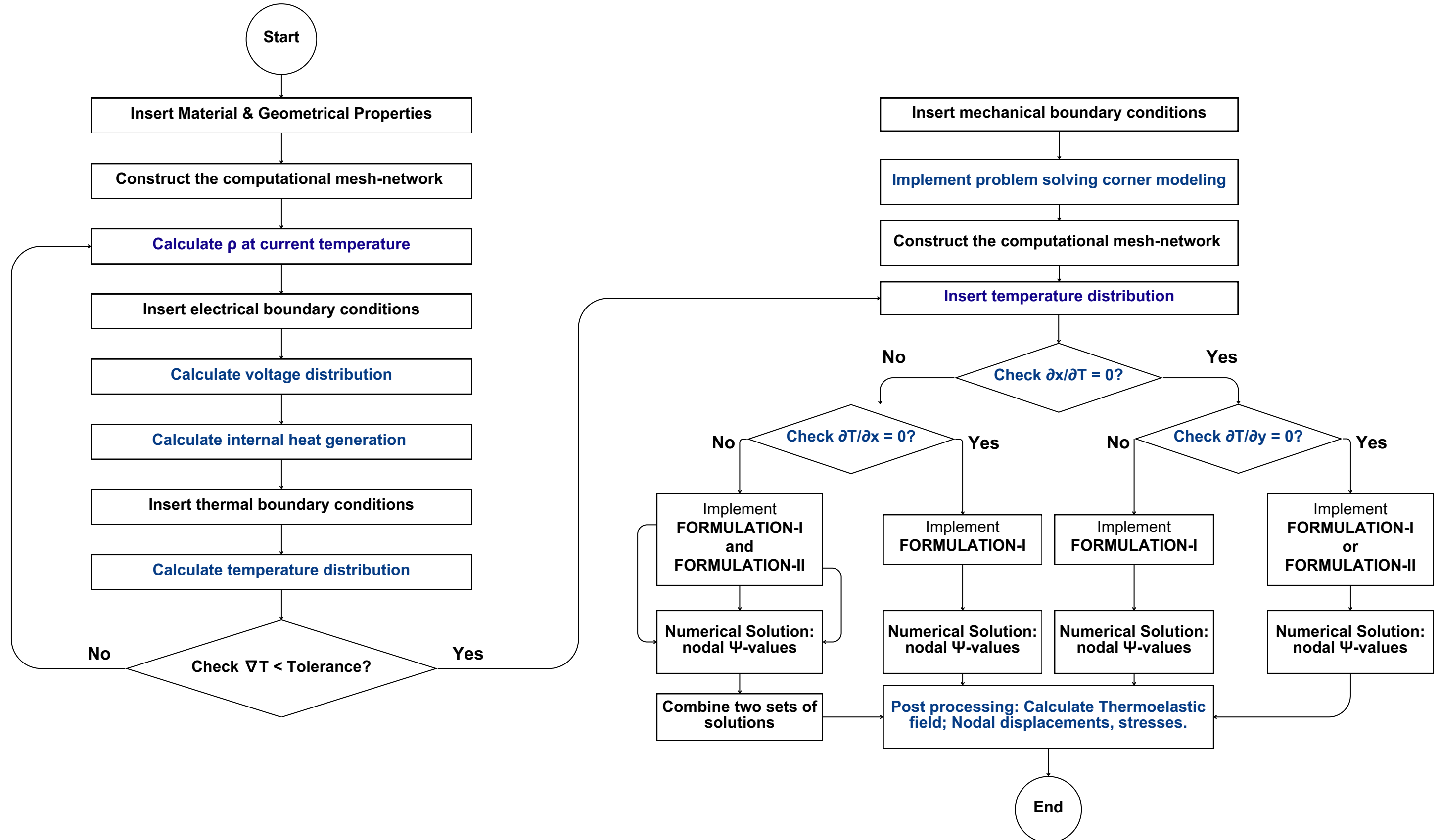
what ensures a **streamlined and efficient solution approach.**”

Research Objectives

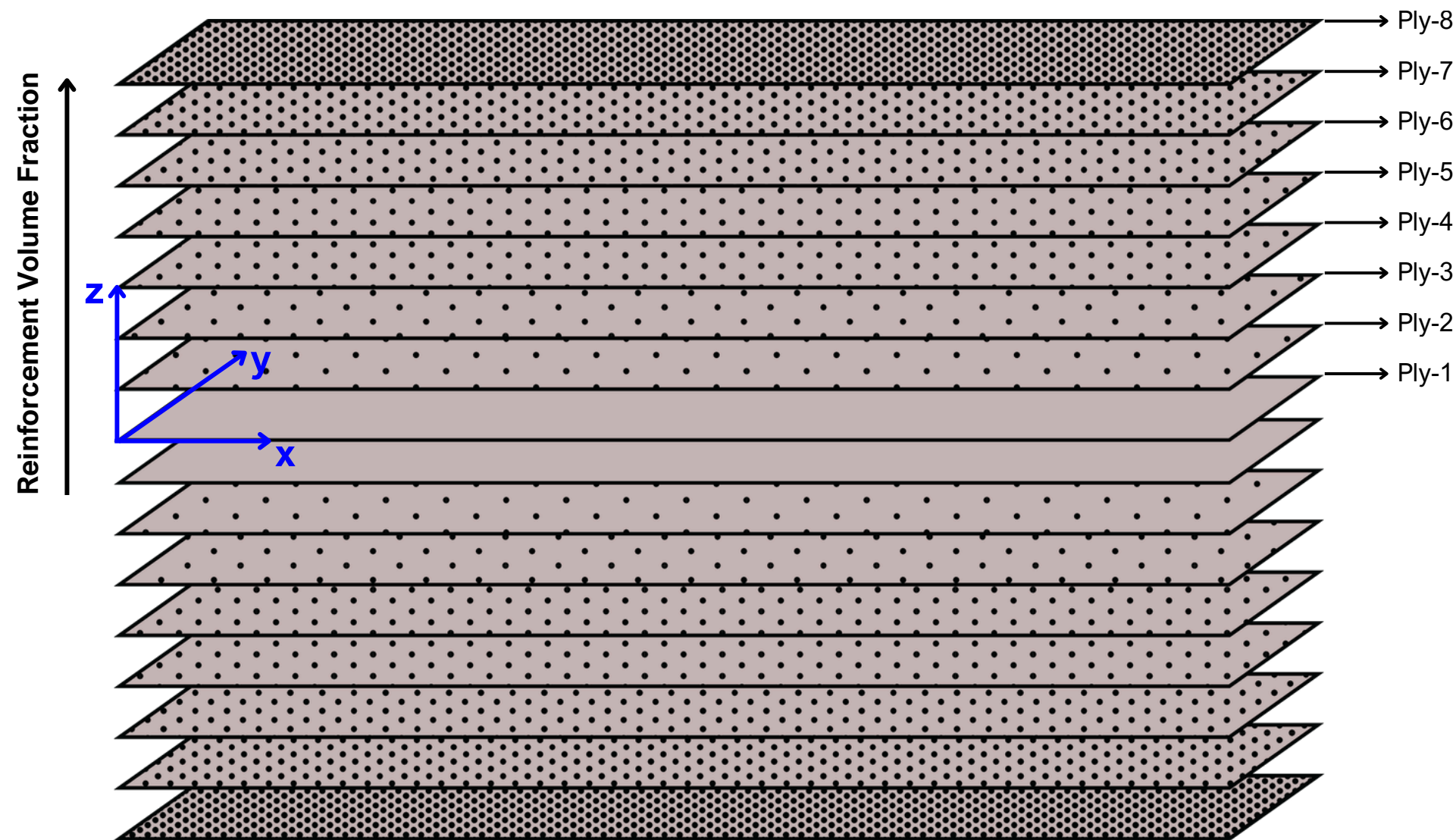
- **Develop a Multiphysics Model for Hybrid Laminated Composites**
 - Implement a semi-analytical approach to analyze electrical, thermal, and mechanical interactions under a DC field.
- **Integrate Temperature-Dependent Material Properties**
 - Incorporate temperature-dependent thermal and electrical conductivity for more realistic simulations.
- **Solve the Coupled Multiphysics Problem Efficiently**
 - Use an iterative approach to couple electrical, thermal, and mechanical domains while employing the Elimination Method for mechanical analysis and the Displacement Function Method for computational efficiency.
- **Validate the Model and Investigate Key Parameters**
 - Compare results with FEA simulations to assess accuracy.
 - Study the effect of changing different key parameters and layer configuration on performance.
- **Provide Insights for Composite Design Optimization**
 - Offer guidelines for optimizing composite design to enhance performance and reliability.

Methodology

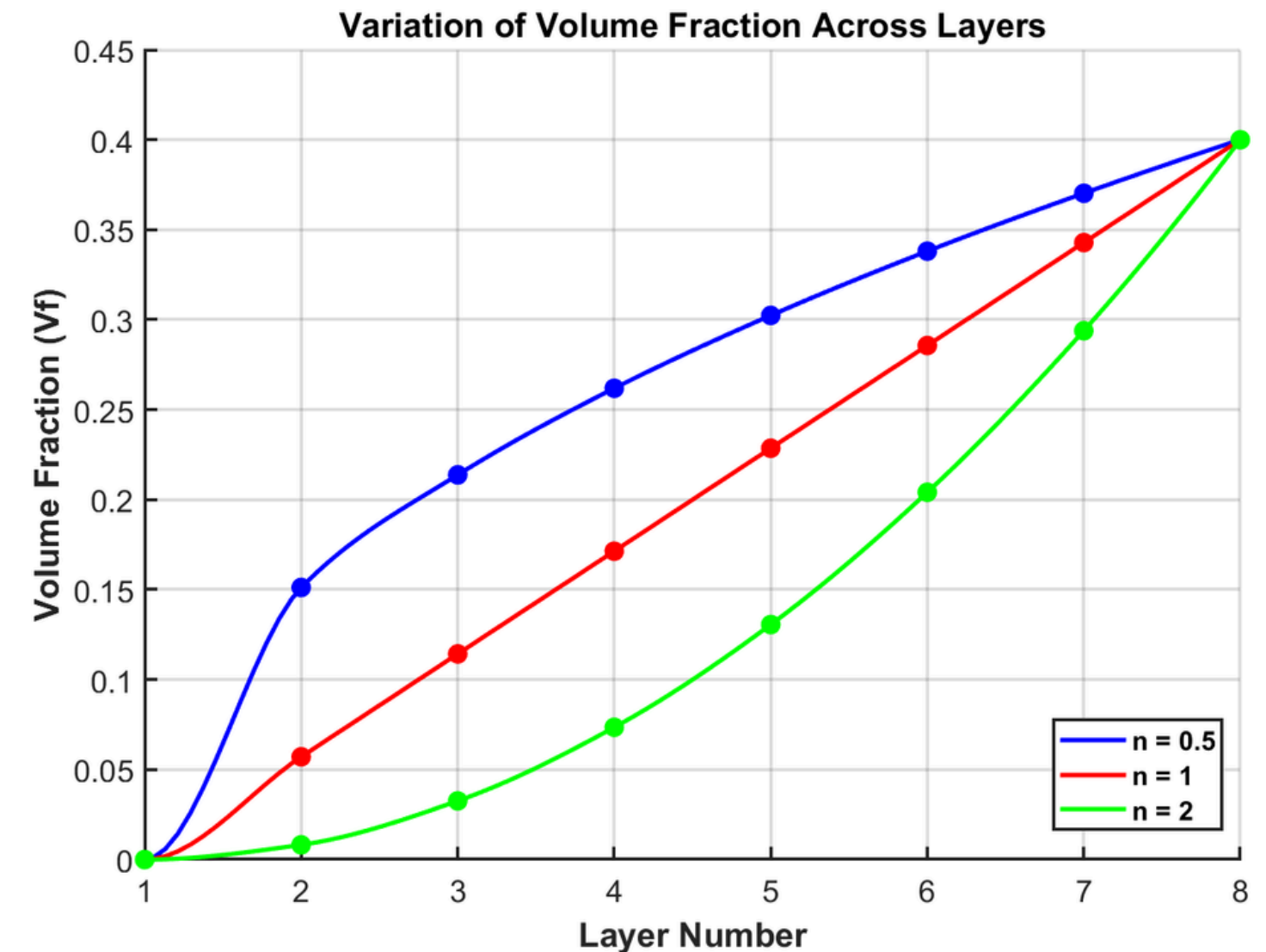
Solution Procedure



The Hybrid Laminated Composite Conductor



- Ply Thickness: 50 μm
- Number of plies: 2 x 8 symmetric layers
- Overall Thickness: 0.8 mm



- Layer-wise Volume Fraction Variation: Defined by $V_f = m \cdot x^n$, where x is the layer number, n is a chosen constant, and m is a scaling factor.
- Gradual Increase: V_f increases from **0** at the midline layer to **0.4** at the outermost layers.

Modeling of the Electrical Field

The maximum and minimum value for electric resistivity of Metal Matrix Composite is given by

$$\rho_{\max} = V_m \rho_m + V_p \rho_p \quad \frac{1}{\rho_{\min}} = \frac{V_m}{\rho_m} + \frac{V_p}{\rho_p}$$

Electrical resistivity varies approximately linearly with temperature and can be expressed as

$$\rho(T) = \rho_0 [1 + \alpha(T - T_0)]$$

Using **Ohm's law**, electric current density resultants for particulate composite laminate are expressed as

$$\begin{bmatrix} J_x \\ J_y \end{bmatrix} = \sum_{k=1}^n \int_{h_{k-1}}^{h_k} \begin{bmatrix} \kappa & 0 \\ 0 & \kappa \end{bmatrix} \begin{bmatrix} \frac{\partial \varphi}{\partial x} \\ \frac{\partial \varphi}{\partial y} \end{bmatrix} dz$$

*Voltage, φ is the governing variable

Considering a constant current density through the uniform cross-section, the **governing equation for the electrical problem** is

$$\nabla \cdot \nabla \varphi = \frac{1}{h} H \frac{\partial^2 \varphi}{\partial x^2} + \frac{1}{h} H \frac{\partial^2 \varphi}{\partial y^2} = 0$$

Simplifying,

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} = 0$$

Joule heating, G can be written in terms of the laminate's conductivity and the electric potential gradient,

$$G = \frac{1}{h} H \left(\frac{\partial \varphi}{\partial x} \right)^2 + \frac{1}{h} H \left(\frac{\partial \varphi}{\partial y} \right)^2$$

Modeling of the Thermal Field

Similarly, **the steady state heat conduction-convection equation with an internal heat source/sink** for a symmetric particulate composite laminate is given by

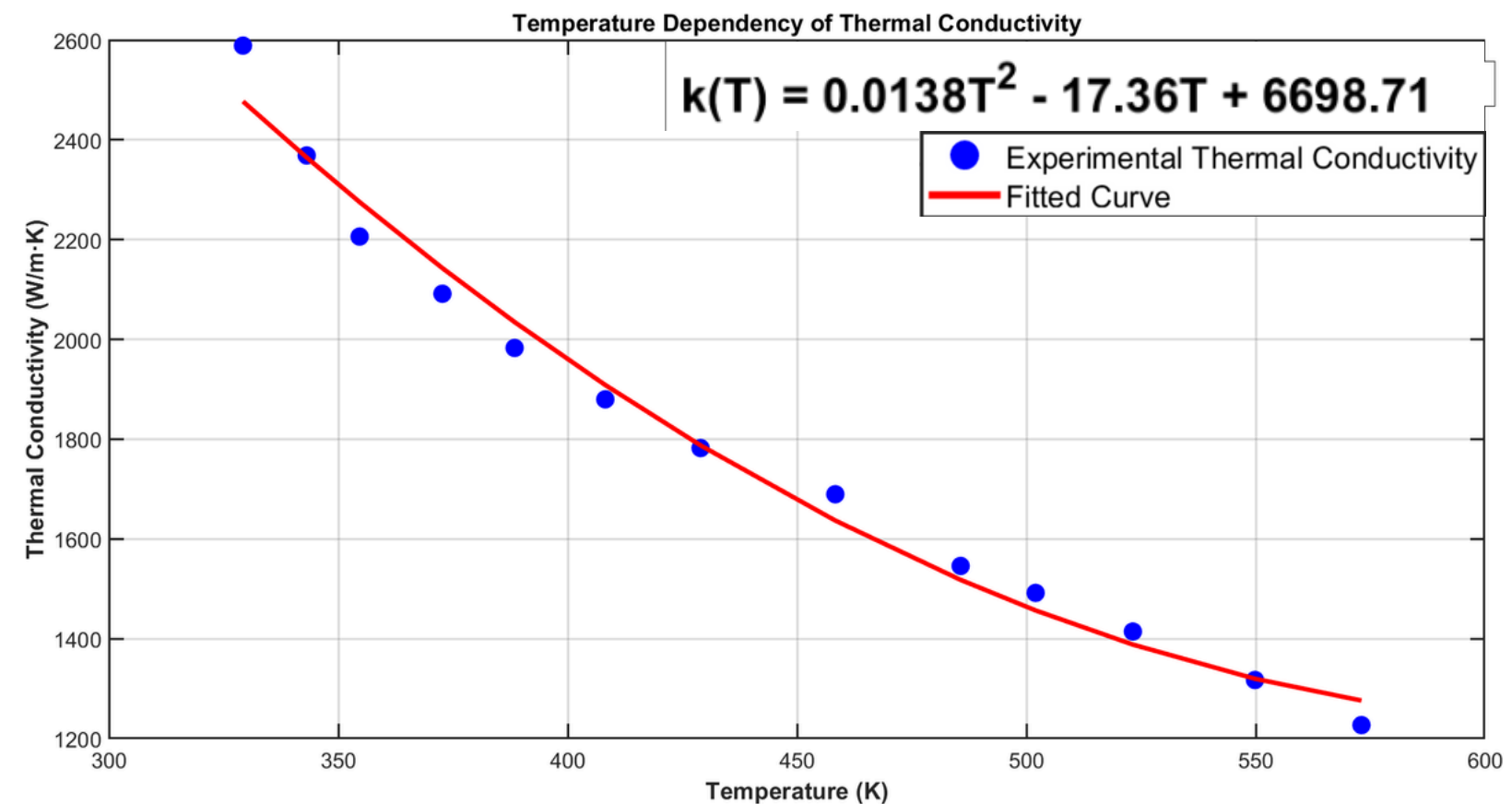
$$\frac{1}{h} H' \frac{\partial^2 T}{\partial x^2} + \frac{1}{h} H' \frac{\partial^2 T}{\partial y^2} + G(x, y) - \frac{2}{h} h_c (T - T_s) = 0$$

The **inward heat flux**, representing heat transfer into the domain, is defined using a Neumann boundary condition

$$\dot{q} = K_c \frac{\partial T}{\partial y}$$

The **temperature-dependent behavior of thermal conductivity** is captured by fitting the experimental data into a quadratic function of the form

$$k(T) = aT^2 + bT + c$$



Graphene's thermal conductivity fitted into quadratic function for modelling

Modeling of the Stress Field

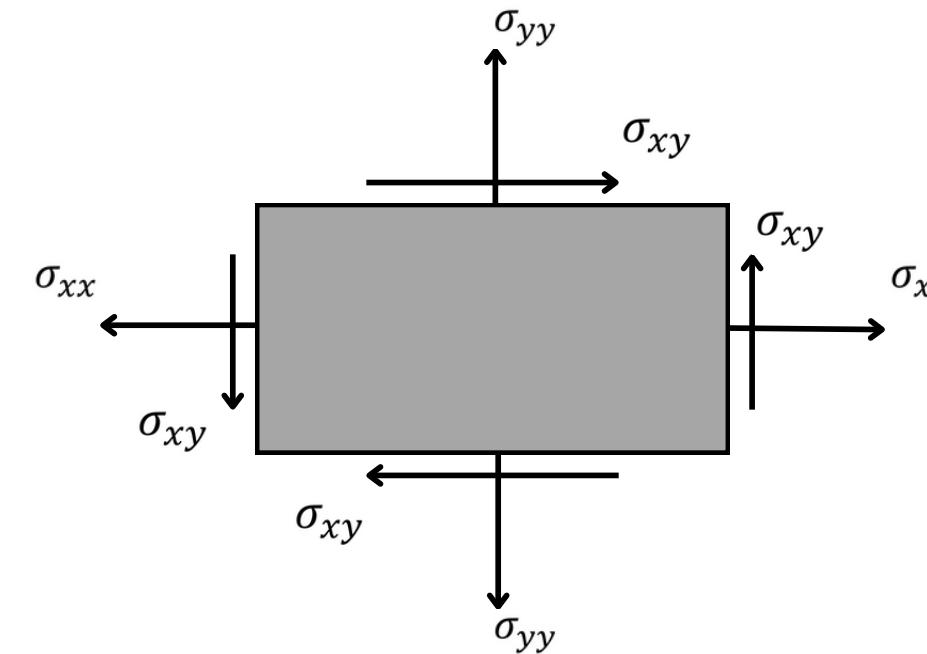
Stress-strain relationship for overall laminate of symmetric isotropic composite with thermal response:

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{bmatrix} = \frac{1}{h} \left(\begin{bmatrix} A_{11} & A_{12} & 0 \\ A_{12} & A_{11} & 0 \\ 0 & 0 & A_{66} \end{bmatrix} \begin{bmatrix} \varepsilon^o_{xx} \\ \varepsilon^o_{yy} \\ \gamma^o_{xy} \end{bmatrix} - \Delta T \begin{bmatrix} \hat{N}_1 \\ \hat{N}_2 \\ 0 \end{bmatrix} \right)$$

$$A_{ij} = \sum_{k=1}^n [(C_{ij})]_k (h_k - h_{k-1}); \quad i, j = 1, 2, 6$$

Strain - Displacement relations

$$\varepsilon_{xx} = \frac{\partial u_x}{\partial x}, \quad \varepsilon_{yy} = \frac{\partial u_y}{\partial y}, \quad \gamma_{xy} = \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}$$



Equilibrium equations having no body forces:

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} = 0$$

$$\frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{xy}}{\partial x} = 0$$

The conditions of equilibrium can be expressed in terms of displacement components,

$$A_{11} \frac{\partial^2 u_x}{\partial x^2} + (A_{12} + A_{66}) \frac{\partial^2 u_y}{\partial x \partial y} + A_{66} \frac{\partial^2 u_x}{\partial y^2} - \hat{N}_1 \frac{\partial T}{\partial x} = 0$$

$$A_{66} \frac{\partial^2 u_y}{\partial x^2} + (A_{12} + A_{66}) \frac{\partial^2 u_y}{\partial x \partial y} + A_{11} \frac{\partial^2 u_y}{\partial y^2} - \hat{N}_2 \frac{\partial T}{\partial y} = 0$$

The Displacement Function

A scalar function called the **Displacement Function (ψ)**, is used to express the displacement components in terms of second-order partial derivatives. This approach simplifies solving elasticity problems, particularly in two-dimensional plane stress and plane strain conditions.

$$u_x(x, y) = \xi_1 \frac{\partial^2 \psi}{\partial x^2} + \xi_2 \frac{\partial^2 \psi}{\partial x \partial y} + \xi_3 \frac{\partial^2 \psi}{\partial y^2}$$

$$u_y(x, y) = \xi_4 \frac{\partial^2 \psi}{\partial x^2} + \xi_5 \frac{\partial^2 \psi}{\partial x \partial y} + \xi_6 \frac{\partial^2 \psi}{\partial y^2}$$

Where ξ s are material-dependent coefficients that relate the potential function to the displacement components.

Displacement Function Modeling

Equilibrium equations in terms of Displacement Function,

$$(\xi_1 A_{11}) \frac{\partial^4 \psi}{\partial x^4} + \{\xi_2 A_{11} + \xi_4 (A_{12} + A_{66})\} \frac{\partial^4 \psi}{\partial x^3 \partial y} + \{\xi_1 A_{66} + \xi_3 A_{11} + \xi_5 (A_{12} + A_{66})\} \frac{\partial^4 \psi}{\partial x^2 \partial y^2} \\ + \{\xi_2 A_{66} + \xi_6 (A_{12} + A_{66})\} \frac{\partial^4 \psi}{\partial x \partial y^3} + (\xi_3 A_{66}) \frac{\partial^4 \psi}{\partial y^4} - \widehat{N}_1 \frac{\partial T}{\partial x} = 0$$

$$(\xi_1 A_{11}) \frac{\partial^4 \psi}{\partial x^4} + \{\xi_1 (A_{12} + A_{66}) + \xi_5 A_{66}\} \frac{\partial^4 \psi}{\partial x^3 \partial y} + \{\xi_2 (A_{12} + A_{66}) + \xi_4 A_{11} + \xi_6 A_{66}\} \frac{\partial^4 \psi}{\partial x^2 \partial y^2} \\ + \{\xi_3 (A_{12} + A_{66}) + \xi_5 A_{11}\} \frac{\partial^4 \psi}{\partial x \partial y^3} + (\xi_3 A_{11}) \frac{\partial^4 \psi}{\partial y^4} - \widehat{N}_2 \frac{\partial T}{\partial y} = 0$$

Formulation - I

$$\frac{\partial T}{\partial x} \neq 0 \quad \frac{\partial T}{\partial y} = 0$$

$$\xi_1 = -\frac{A_{66}}{A_{12} + A_{66}} \quad \xi_1 = -\frac{A_{11}}{A_{12} + A_{66}}$$

$$\xi_2 = \xi_4 = \xi_6 = 0 \quad \xi_5 = 1$$

$$\frac{\partial^4 \psi}{\partial x^4} + \left(\frac{A_{11}^2 - A_{12}^2 - 2A_{12}A_{66}}{A_{11}A_{66}} \right) \frac{\partial^4 \psi}{\partial x^2 \partial y^2} + \frac{\partial^4 \psi}{\partial y^4} = - \left(\frac{A_{12} + A_{66}}{A_{11}A_{66}} \right) \widehat{N}_1 \frac{\partial T}{\partial x}$$

Formulation - II

$$\frac{\partial T}{\partial x} = 0 \quad \frac{\partial T}{\partial y} \neq 0$$

$$\xi_4 = -\frac{A_{11}}{A_{12} + A_{66}} \quad \xi_6 = -\frac{A_{66}}{A_{12} + A_{66}}$$

$$\xi_1 = \xi_3 = \xi_5 = 0 \quad \xi_2 = 1$$

$$\frac{\partial^4 \psi}{\partial x^4} + \left(\frac{A_{11}^2 - A_{12}^2 - 2A_{12}A_{66}}{A_{11}A_{66}} \right) \frac{\partial^4 \psi}{\partial x^2 \partial y^2} + \frac{\partial^4 \psi}{\partial y^4} = - \left(\frac{A_{12} + A_{66}}{A_{11}A_{66}} \right) \widehat{N}_2 \frac{\partial T}{\partial y}$$

Displacement Function Modeling

The mechanical parameters are calculated from the value of ψ ,

Displacements:

$$u_x(x, y) = -\left(\frac{A_{66}}{A_{12} + A_{66}}\right) \frac{\partial^2 \psi}{\partial x^2} - \left(\frac{A_{11}}{A_{12} + A_{66}}\right) \frac{\partial^2 \psi}{\partial y^2}$$
$$u_y(x, y) = \frac{\partial^2 \psi}{\partial x \partial y}$$

Stresses:

$$\sigma_{xx}(x, y) = \frac{1}{h} \left[\left(\frac{A_{12}^2 + A_{12}A_{66} - A_{11}^2}{A_{12} + A_{66}} \right) \frac{\partial^3 \psi}{\partial x \partial y^2} - \left(\frac{A_{11}A_{66}}{A_{12} + A_{66}} \right) \frac{\partial^3 \psi}{\partial x^3} - \hat{N}_1 \Delta T \right]$$
$$\sigma_{yy}(x, y) = \frac{1}{h} \left[\left(\frac{A_{11}A_{66}}{A_{12} + A_{66}} \right) \frac{\partial^3 \psi}{\partial x \partial y^2} - \left(\frac{A_{12}A_{66}}{A_{12} + A_{66}} \right) \frac{\partial^3 \psi}{\partial x^3} - \hat{N}_2 \Delta T \right]$$
$$\sigma_{xy}(x, y) = \frac{1}{h} \left[\left(-\frac{A_{11}A_{66}}{A_{12} + A_{66}} \right) \frac{\partial^3 \psi}{\partial y^3} + \left(\frac{A_{12}A_{66}}{A_{12} + A_{66}} \right) \frac{\partial^3 \psi}{\partial x^2 \partial y} \right]$$

Strains:

$$\varepsilon_{xx}(x, y) = -\left(\frac{A_{66}}{A_{12} + A_{66}}\right) \frac{\partial^3 \psi}{\partial x^3} - \left(\frac{A_{11}}{A_{12} + A_{66}}\right) \frac{\partial^3 \psi}{\partial x \partial y^2}$$
$$\varepsilon_{yy}(x, y) = \frac{\partial^3 \psi}{\partial x \partial y^2}$$
$$\gamma_{xy}(x, y) = -\left(\frac{A_{11}}{A_{12} + A_{66}}\right) \frac{\partial^3 \psi}{\partial y^3} + \left(\frac{A_{12}}{A_{12} + A_{66}}\right) \frac{\partial^3 \psi}{\partial x^2 \partial y}$$

Only for Formulation-I is shown.
Formulation-II is similar.

Discretization of Governing Equations

- The discrete form of the governing equation for the electrical problem:

$$h_y^2(\varphi_{i+1,j} - 2\varphi_{i,j} + \varphi_{i-1,j}) + h_x^2(\varphi_{i,j+1} - 2\varphi_{i,j} + \varphi_{i,j-1}) = 0$$

- The simplified discrete form of the governing equation for the thermal problem:

$$T_{i,j} = \frac{\frac{H}{h} \left(\frac{T_{i+1,j} + T_{i-1,j}}{h_x^2} + \frac{T_{i,j+1} + T_{i,j-1}}{h_y^2} \right) + G_{i,j} + \frac{2}{h} h_c T_s}{\frac{2H}{h} \left(\frac{1}{h_x^2} + \frac{1}{h_y^2} \right) + \frac{2}{h} h_c}$$

Discretization of Governing Equations

- The discrete form of the governing equation for the Mechanical problem:

Formulation – I: (Takes care of axial thermal gradient)

$$\begin{aligned}
 &pk1(\psi_{i+2,j} + \psi_{i-2,j}) - (4pk1 + 2pk2)(\psi_{i+1,j} + \psi_{i-1,j}) - (2pk2 + 4pk3)(\psi_{i,j+1} - \psi_{i,j-1}) \\
 &\quad + (6pk1 + 4pk2 + 6pk3)\psi_{i,j} \\
 &\quad + pk2(\psi_{i+1,j+1} + \psi_{i+1,j-1} + \psi_{i-1,j+1} + \psi_{i-1,j-1}) + pk3(\psi_{i,j+2} + \psi_{i,j-2}) \\
 &= pk4(T_{i+1,j} - T_{i-1,j})
 \end{aligned}$$

$$\begin{aligned}
 pk1 &= \frac{1}{h_x^4}, \\
 pk2 &= \frac{1}{h_x^2 h_y^2} \left[\frac{A_{11}^2 - A_{12}^2 - 2A_{12}A_{66}}{A_{11}A_{66}} \right] \\
 pk3 &= \frac{1}{h_y^4} \\
 pk4 &= - \left(\frac{A_{12} + A_{66}}{A_{11}A_{66}} \right) \frac{\hat{N}_1}{2h_x}
 \end{aligned}$$

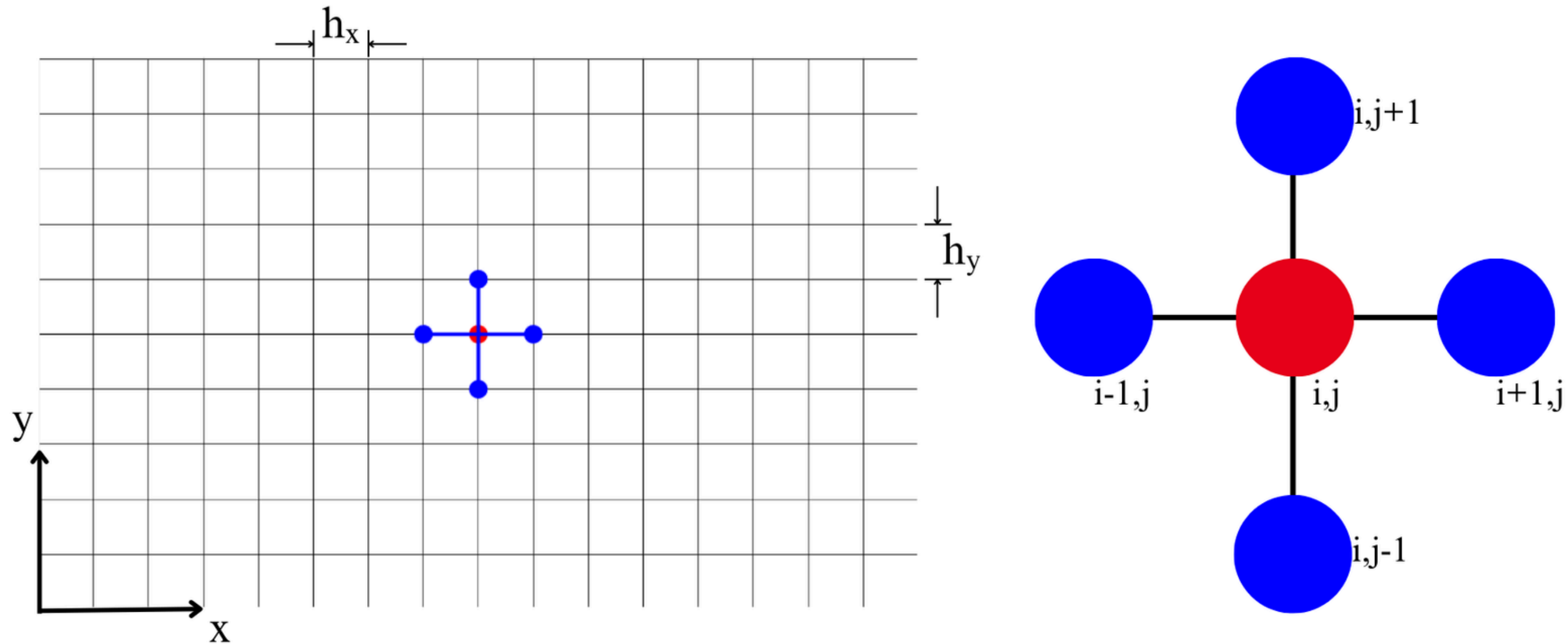
Formulation – II (Takes care of transverse thermal gradient)

$$\begin{aligned}
 &pk1(\psi_{i+2,j} + \psi_{i-2,j}) - (4pk1 + 2pk2)(\psi_{i+1,j} + \psi_{i-1,j}) \\
 &\quad - (2pk2 + 4pk3)(\psi_{i,j+1} - \psi_{i,j-1}) + (6pk1 + 4pk2 + 6pk3)\psi_{i,j} \\
 &\quad + pk2(\psi_{i+1,j+1} + \psi_{i+1,j-1} + \psi_{i-1,j+1} + \psi_{i-1,j-1}) + pk3(\psi_{i,j+2} + \psi_{i,j-2}) \\
 &= pk5(T_{i,j+1} - T_{i,j-1})
 \end{aligned}$$

$$pk5 = - \left(\frac{A_{12} + A_{66}}{A_{11}A_{66}} \right) \frac{\hat{N}_2}{2h_y}$$

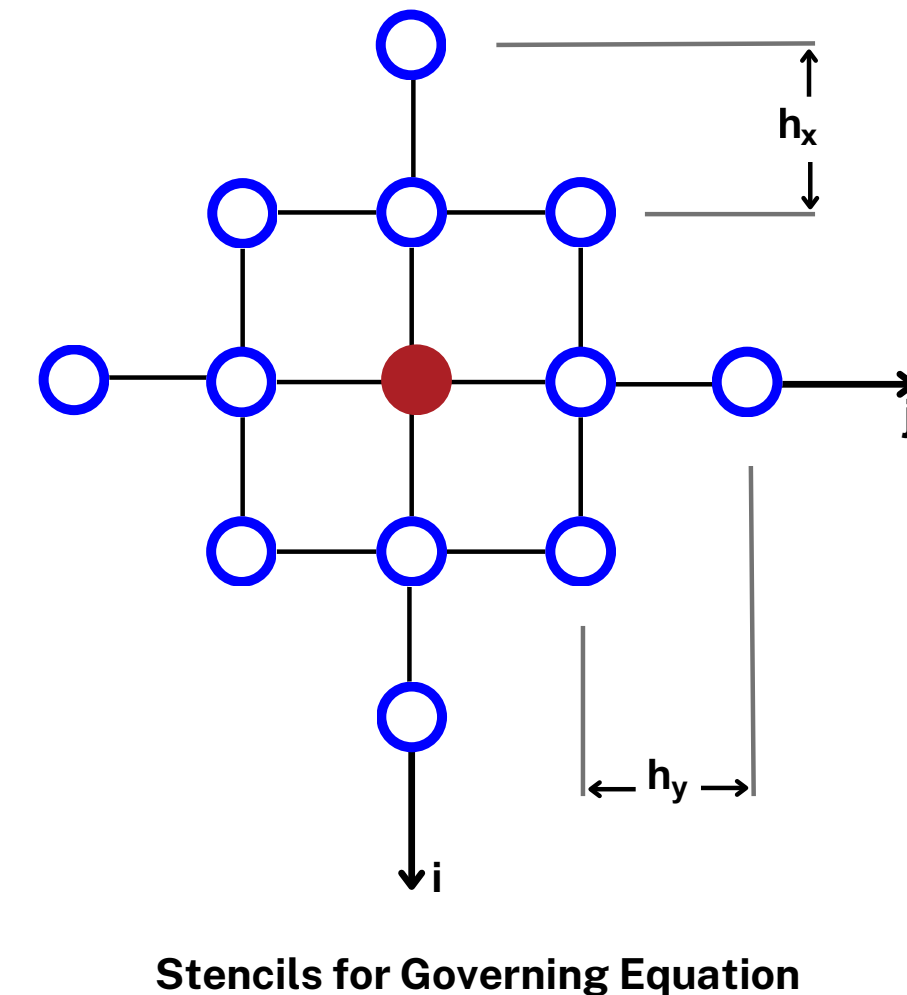
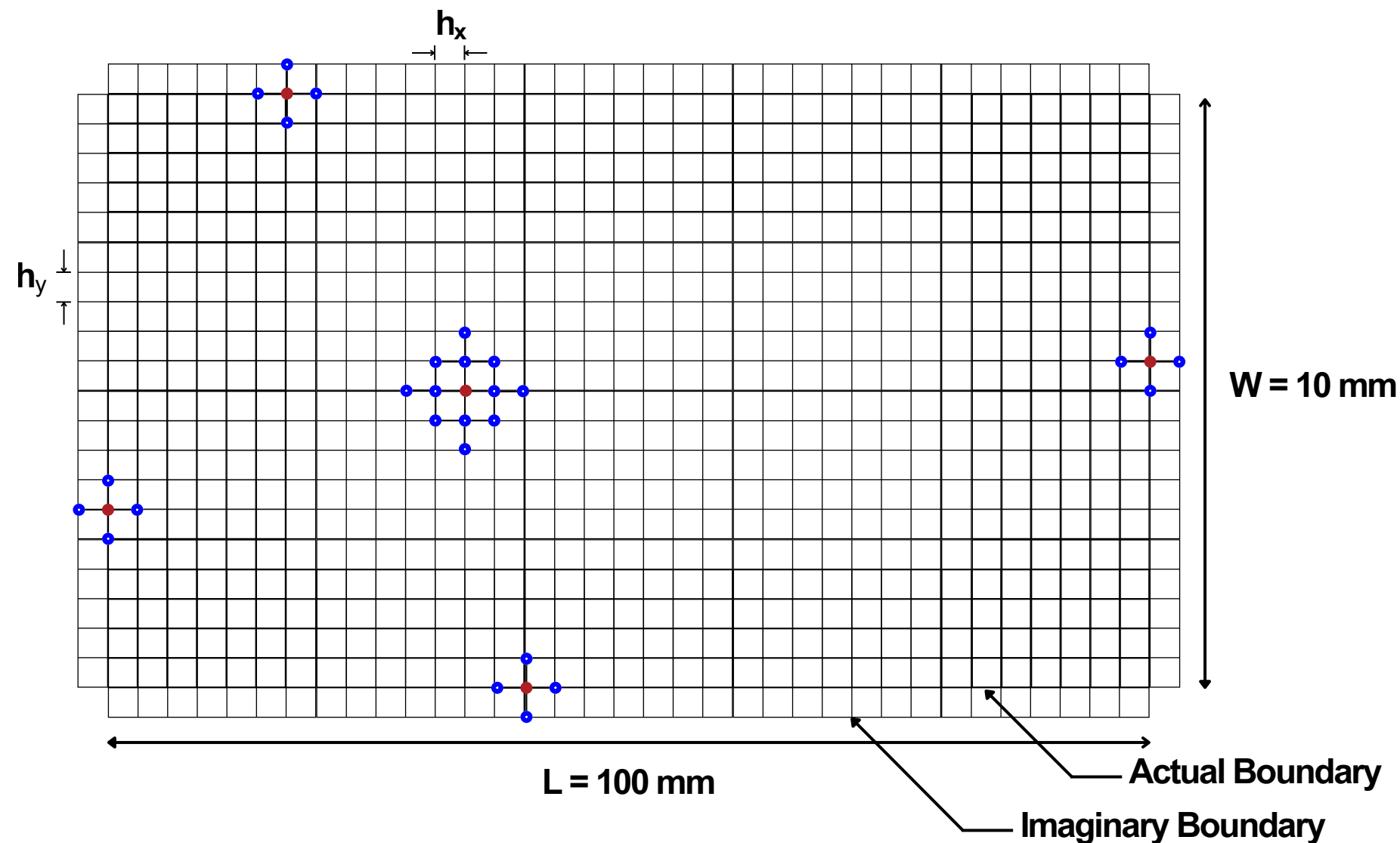
Discretization of Electro-thermal Field

- The computational domain is subdivided into nodes, each representing subdomain properties and field variables, with a **second-order FDM scheme** requiring up to **5 nodal points** for accurate equation representation.



Discretization of Mechanical Problem

- The computational domain is subdivided into nodes, each representing subdomain properties and field variables, with a **fourth-order FDM scheme** requiring up to **13 nodal points** for accurate equation representation.
- Mesh Configuration:** A **401 × 101 nodes** mesh is used for a $100 \times 10 \text{ mm}^2$ domain. Imaginary layer of nodes is taken to accurately model boundaries on the 4th order governing equation.



Mesh Converging Characteristic

Mesh Size	Electro-thermal Response				Stresses at Midpoint			
	Maximum Voltage (mV)	Change (%)	Midpoint Temperature K	Change (%)	σ_{xx} (MPa)	Change (%)	σ_{yy} (MPa)	Change (%)
51 x 11	194.54	-	353.55	-	-43.79	-	-40.20	-
101 x 21	195.89	0.69	355.67	0.60	-45.63	4.20	-42.21	5.00
201 x 41	196.65	0.39	356.76	0.31	-46.56	2.04	-43.18	2.29
301 x 61	196.91	0.13	357.12	0.10	-46.87	0.67	-43.49	0.74
401 x 81	197.04	0.07	357.31	0.05	-47.02	0.32	-43.65	0.37
501 x 101	197.12	0.04	357.41	0.03	-47.12	0.20	-43.74	0.20

After 401 x 81 mesh, it gets **increasingly computationally expensive** for very little improvement in result.

Key Assumptions

- **Symmetric Laminate**

Simplifies the analytical formulation reducing computational cost.

- **No Impurities or Defects**

The materials are assumed to be ideal, without voids or inclusions.

- **Negligible Radiation**

Thermal radiation effects are ignored due to their minimal contribution in this temperature range.

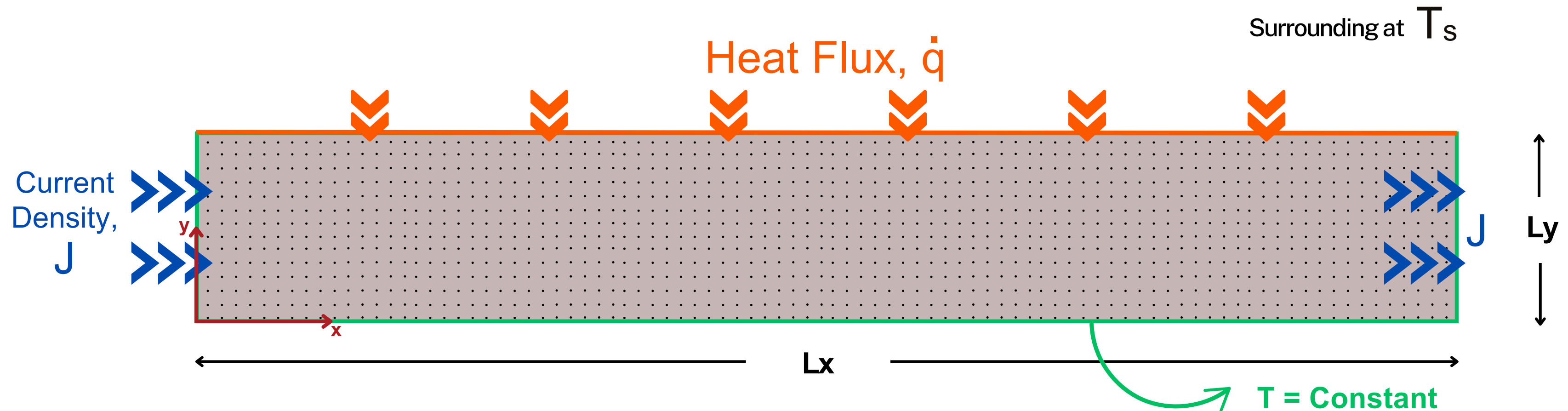
- **No Current Loss**

The system is considered electrically isolated with no leakage currents.

- **Constant Bonding Material Properties**

The bonding layers (e.g., silver paste) are assumed to have temperature-independent properties.

Problem Definition



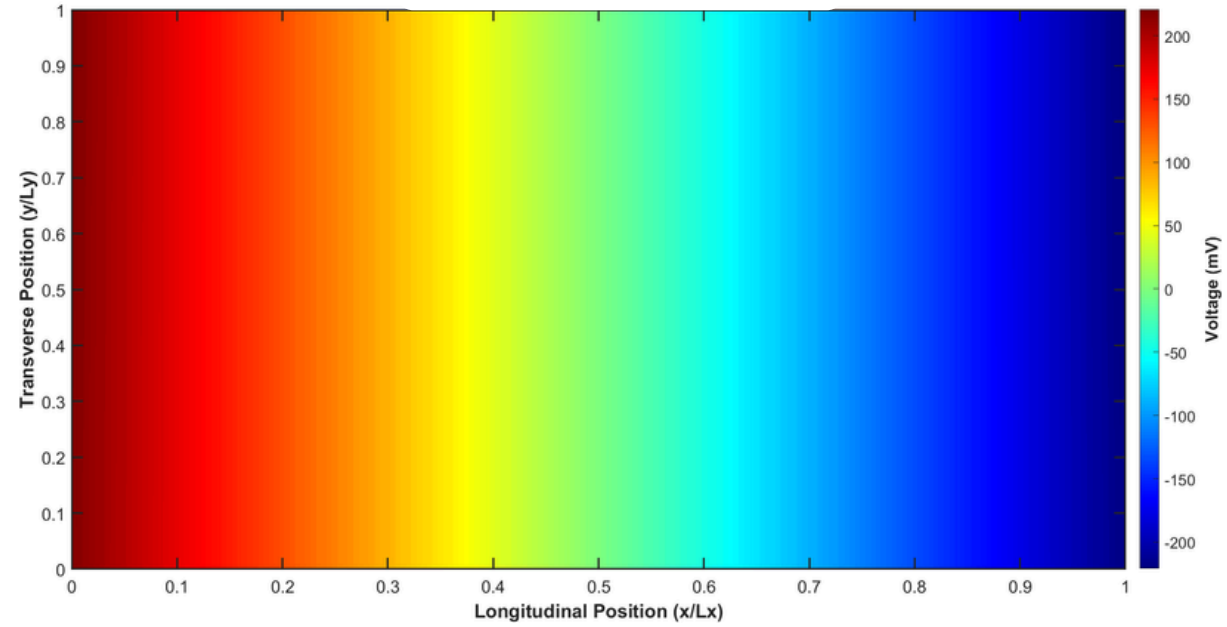
- Electrically insulated top and bottom boundary.
- **Free convection heat transfer** from the uppermost layer.
- Mechanically **Clamped on all 4 sides**.

- Example Problem Materials:
 - **Matrix: Aluminum**
 - **Particle: Silicon Carbide (Si-C)**
 - $\text{Max } V_f = 0.4, n = 0.5$

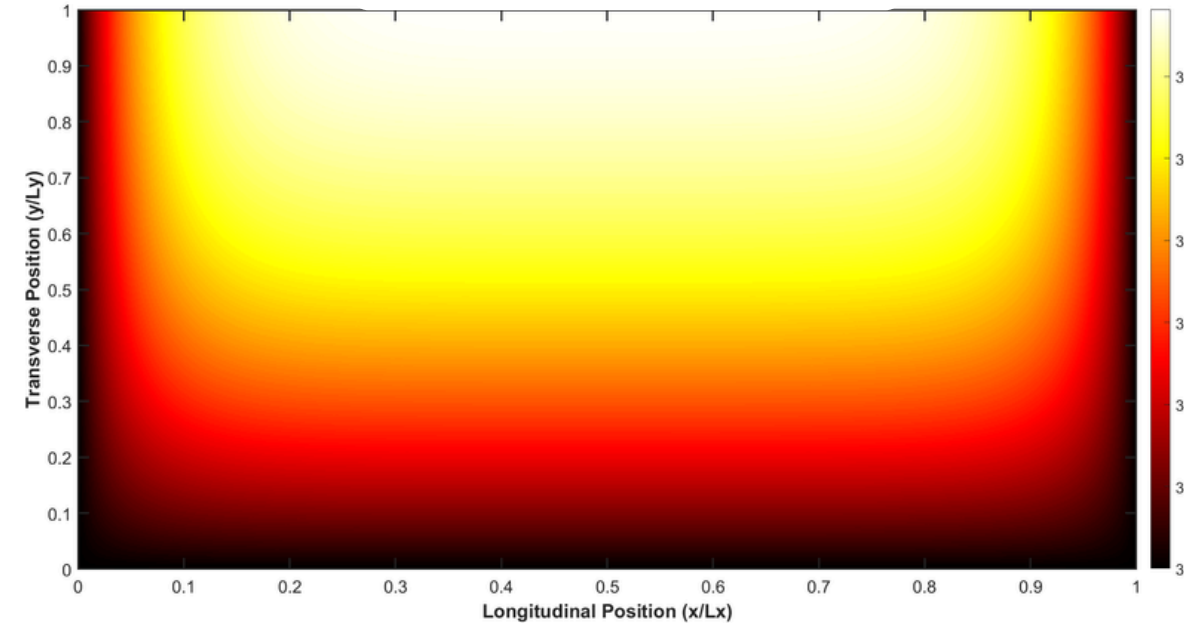
Results

[Al-SiC, $n = 0.5$, $V_f_{\max} = 0.4$, $J = 1E08A/m^2$, $q = 200,000 J/m^2$, $h_c = 15$, $T_s = 315$]

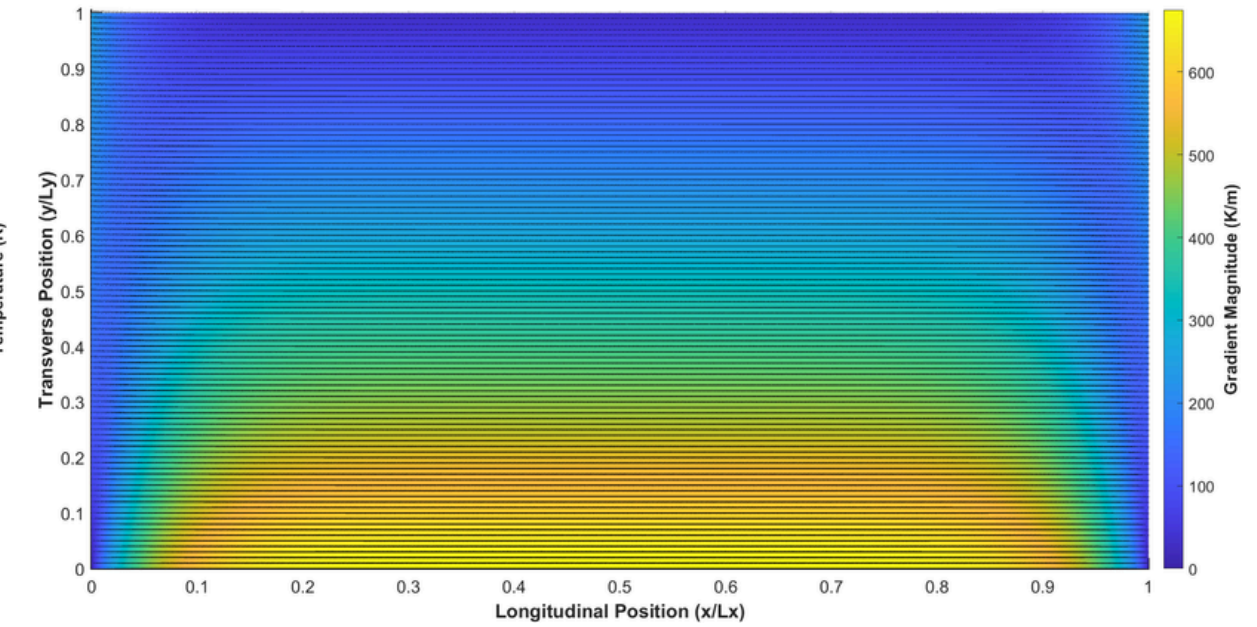
Voltage Distribution



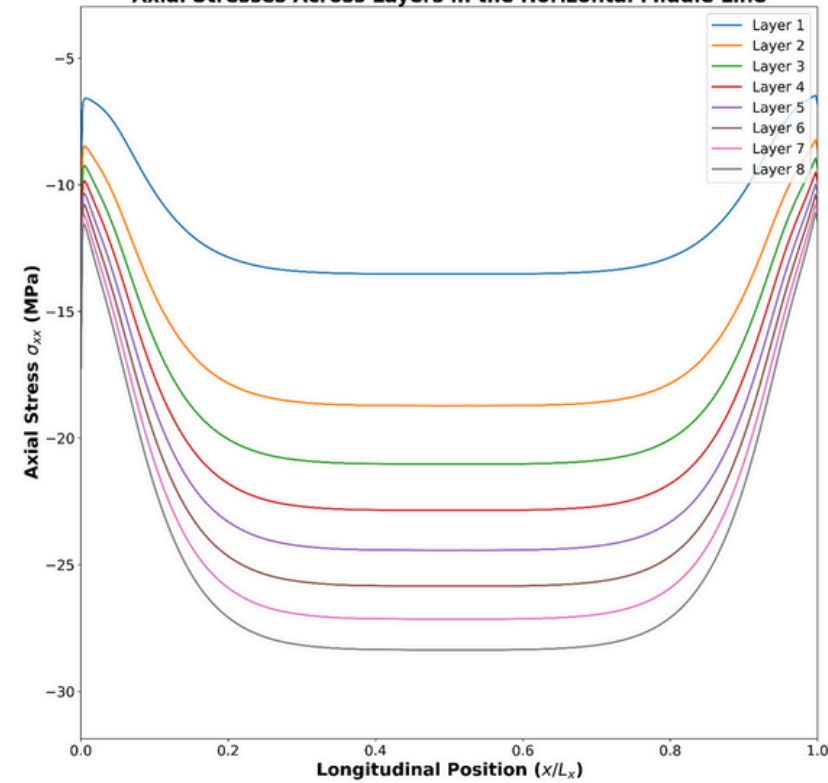
Temperature Distribution



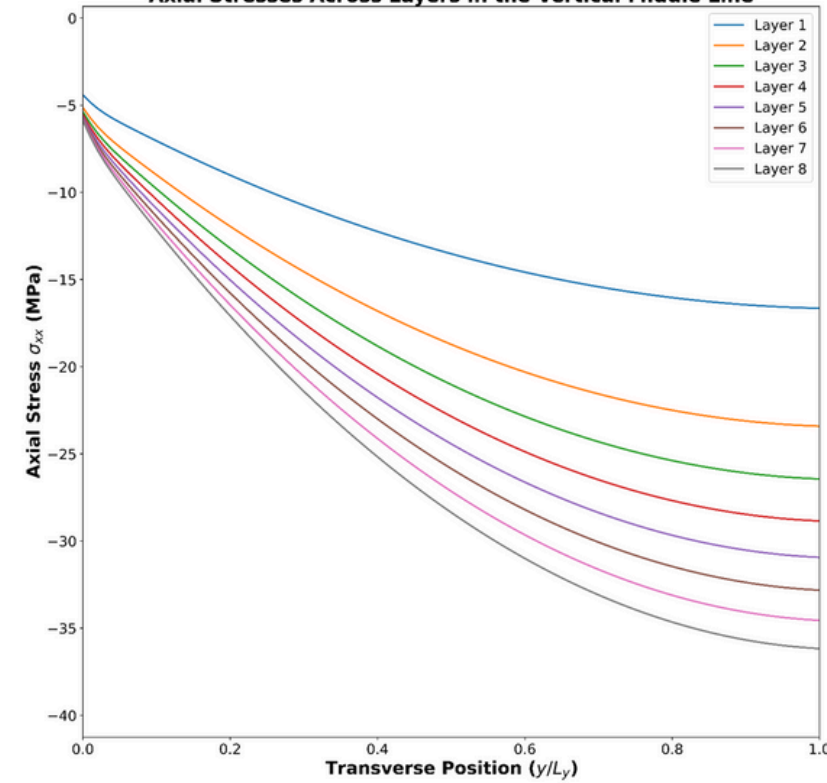
Temperature Gradient Distribution



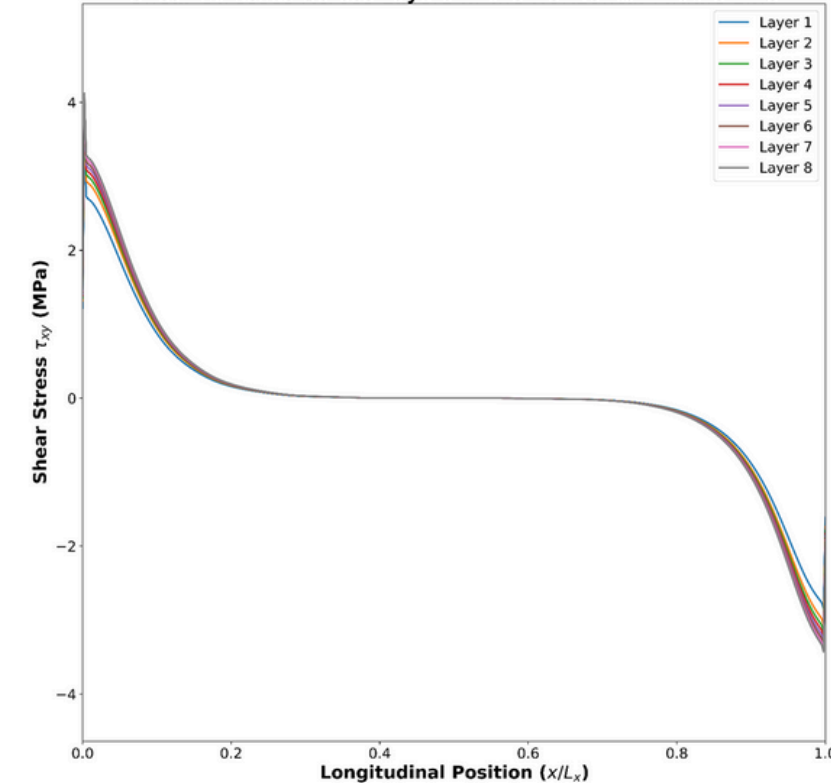
Axial Stresses Across Layers in the Horizontal Middle Line



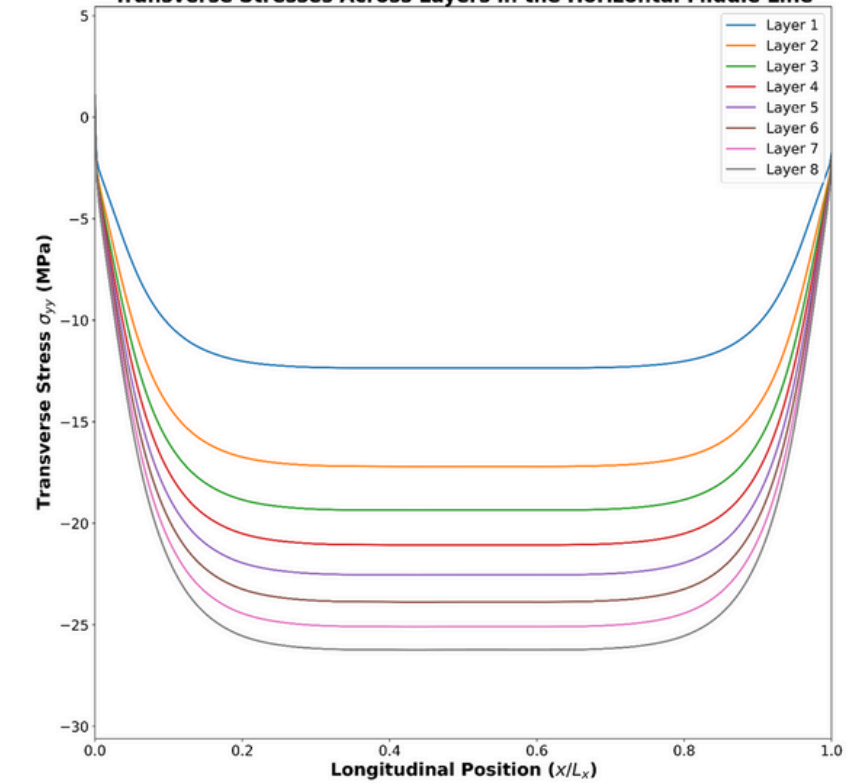
Axial Stresses Across Layers in the Vertical Middle Line



Shear Stresses Across Layers in the Horizontal Middle Line



Transverse Stresses Across Layers in the Horizontal Middle Line



Axial Stress in Longitudinal and Transverse Direction

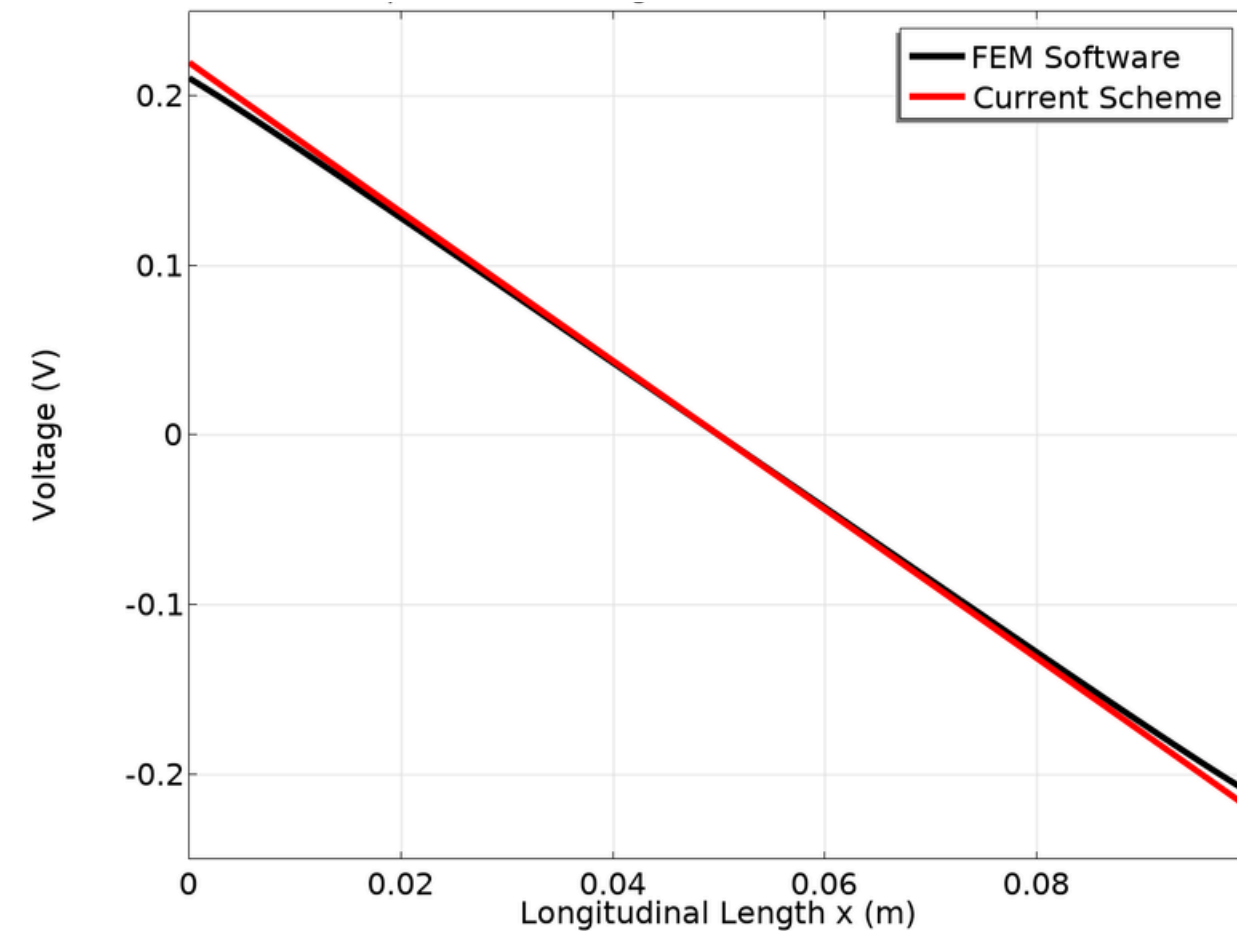
Shear Stress

Transverse Stress

Validation

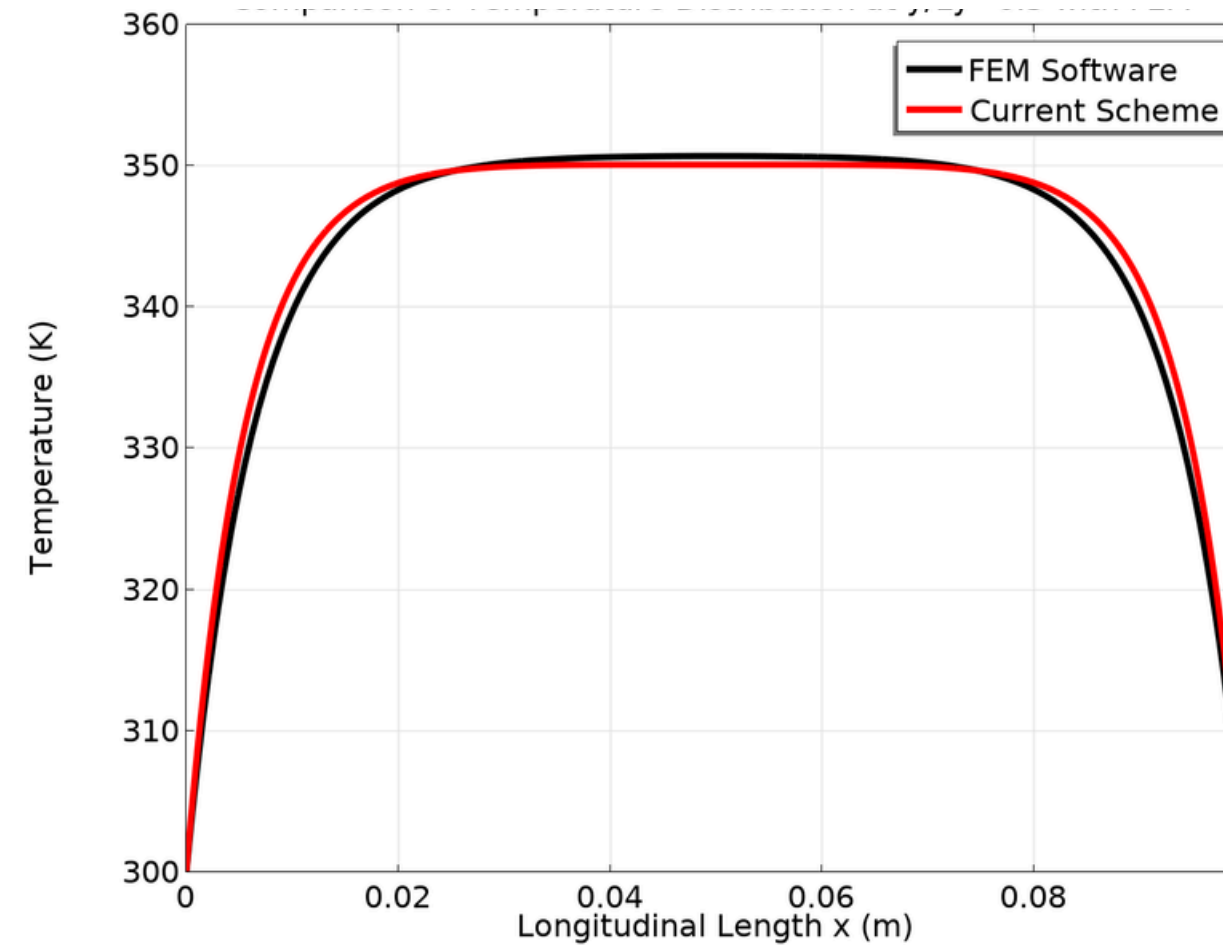
[Al-SiC, $n = 0.5$, $V_{f_max} = 0.4$, $J = 1E08A/m^2$, $q = 200,000 J/m^2$, $h_c = 15$, $T_s = 315$]

Voltage Distribution Comparison



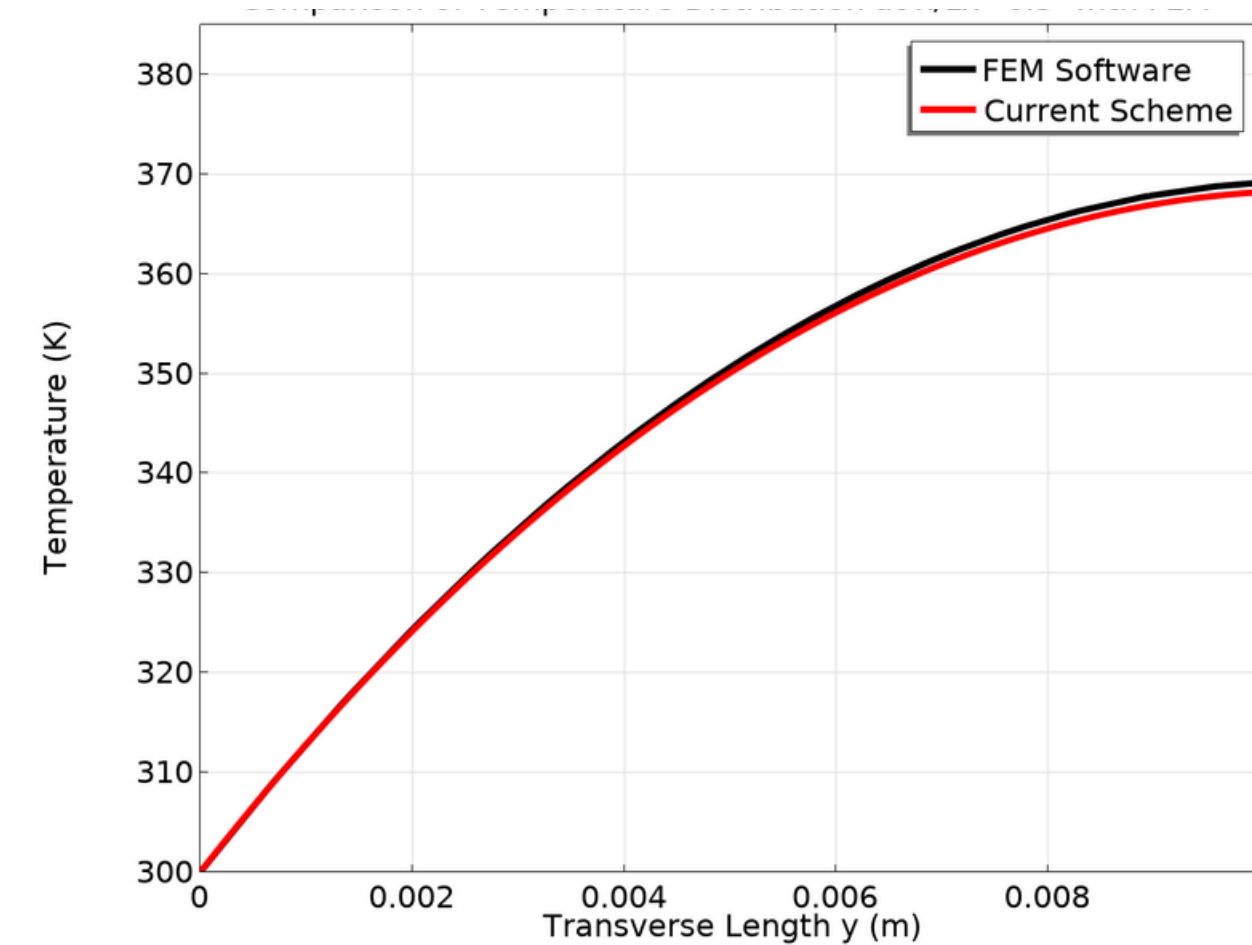
- Voltage Distribution: **Strong agreement** between our scheme and FEM, with minimal deviation.

Axial Temperature Profile Comparison



- Axial Temperature Distribution: **Good agreement**, slight deviations near peak values.

Transverse Temperature Profile Comparison

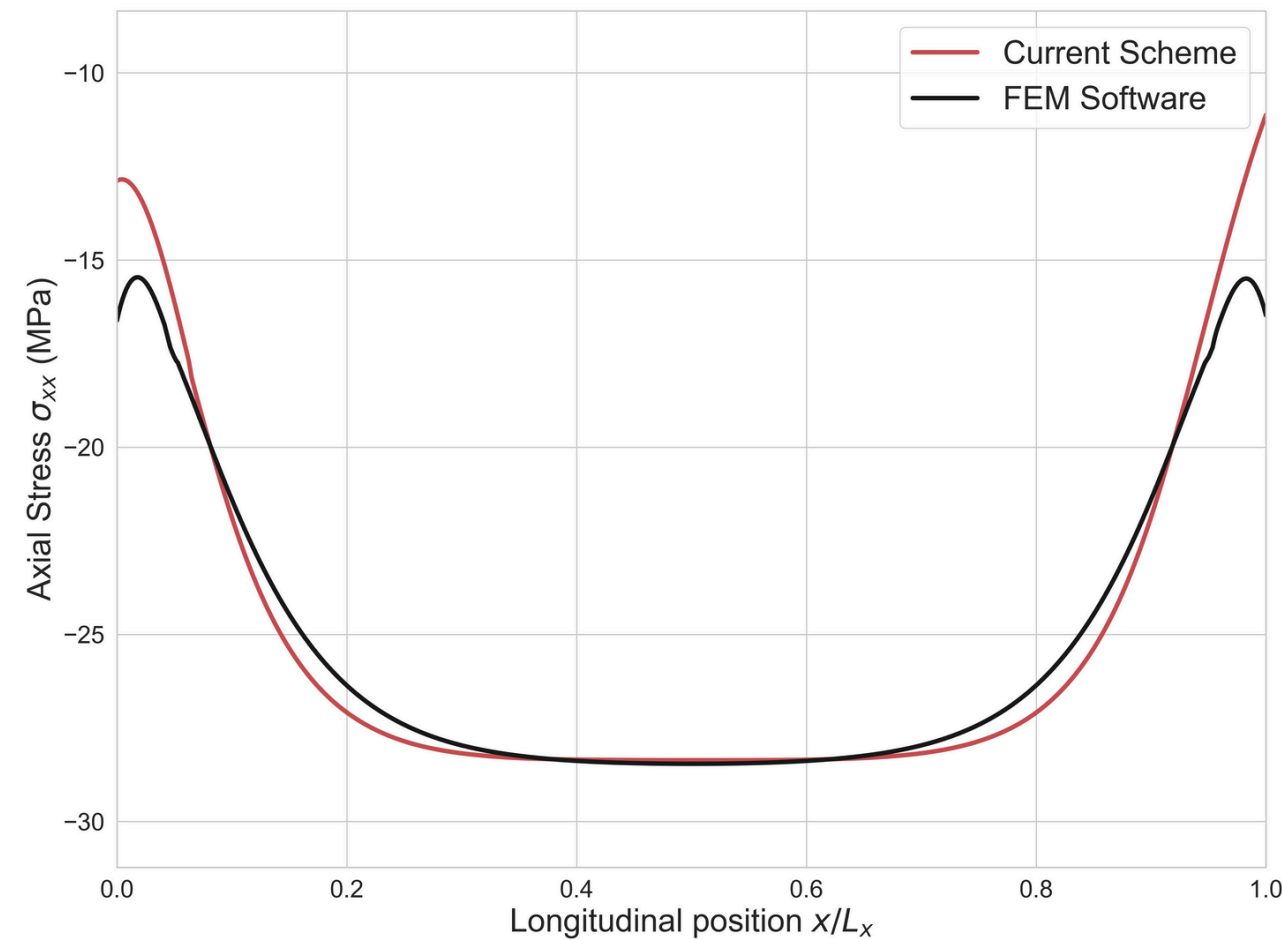


- Transverse Temperature Distribution: **Close match**, minor differences at higher temperatures.

Validation

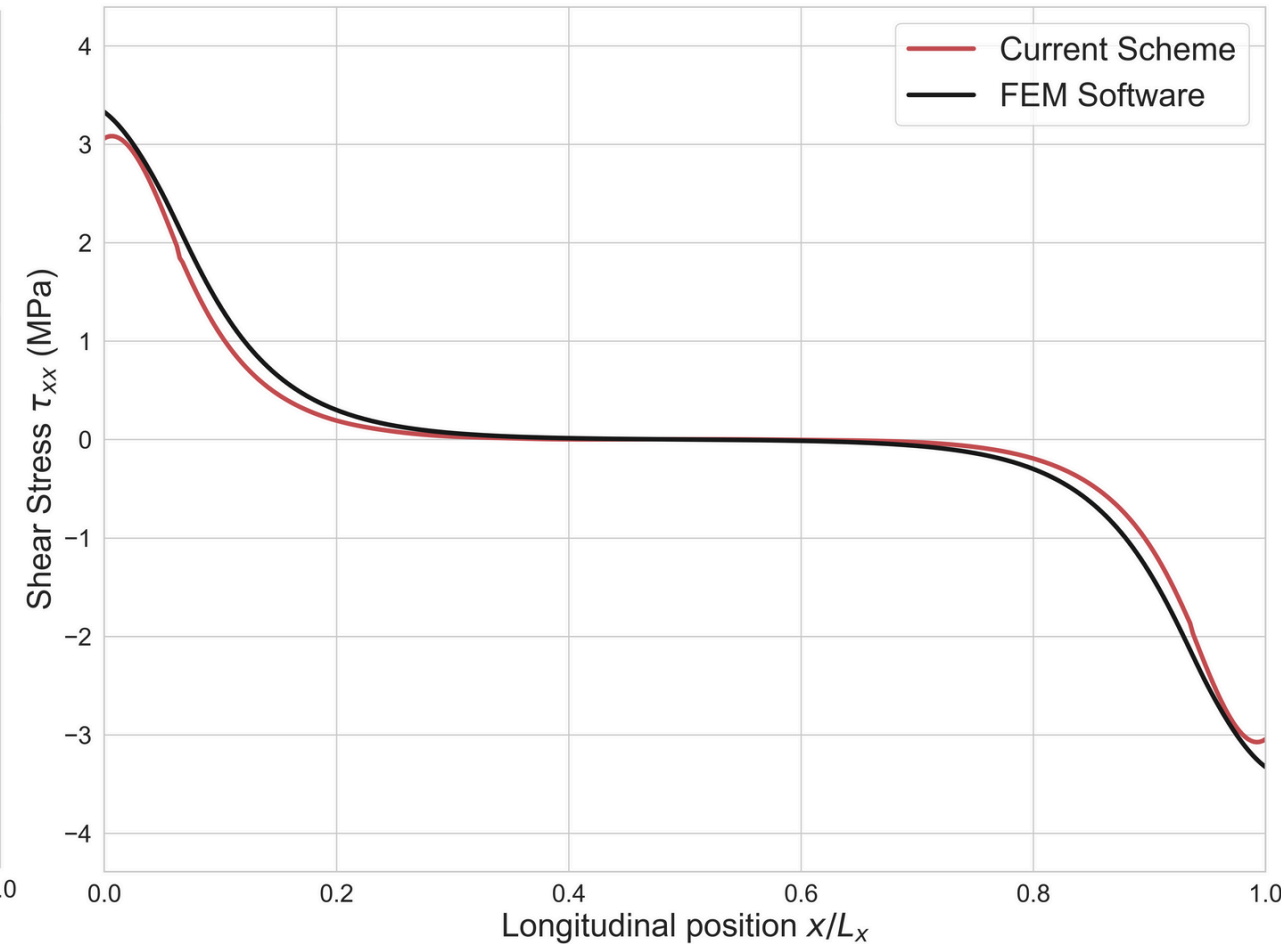
[Al-SiC, $n = 0.5$, $Vf_{max} = 0.4$, $J = 1E08A/m^2$, $q = 200,000 J/m^2$, $hc = 15$, $T_s = 315$]

Comparison of Axial stress at $y/L_y = 0.5$ with FEM



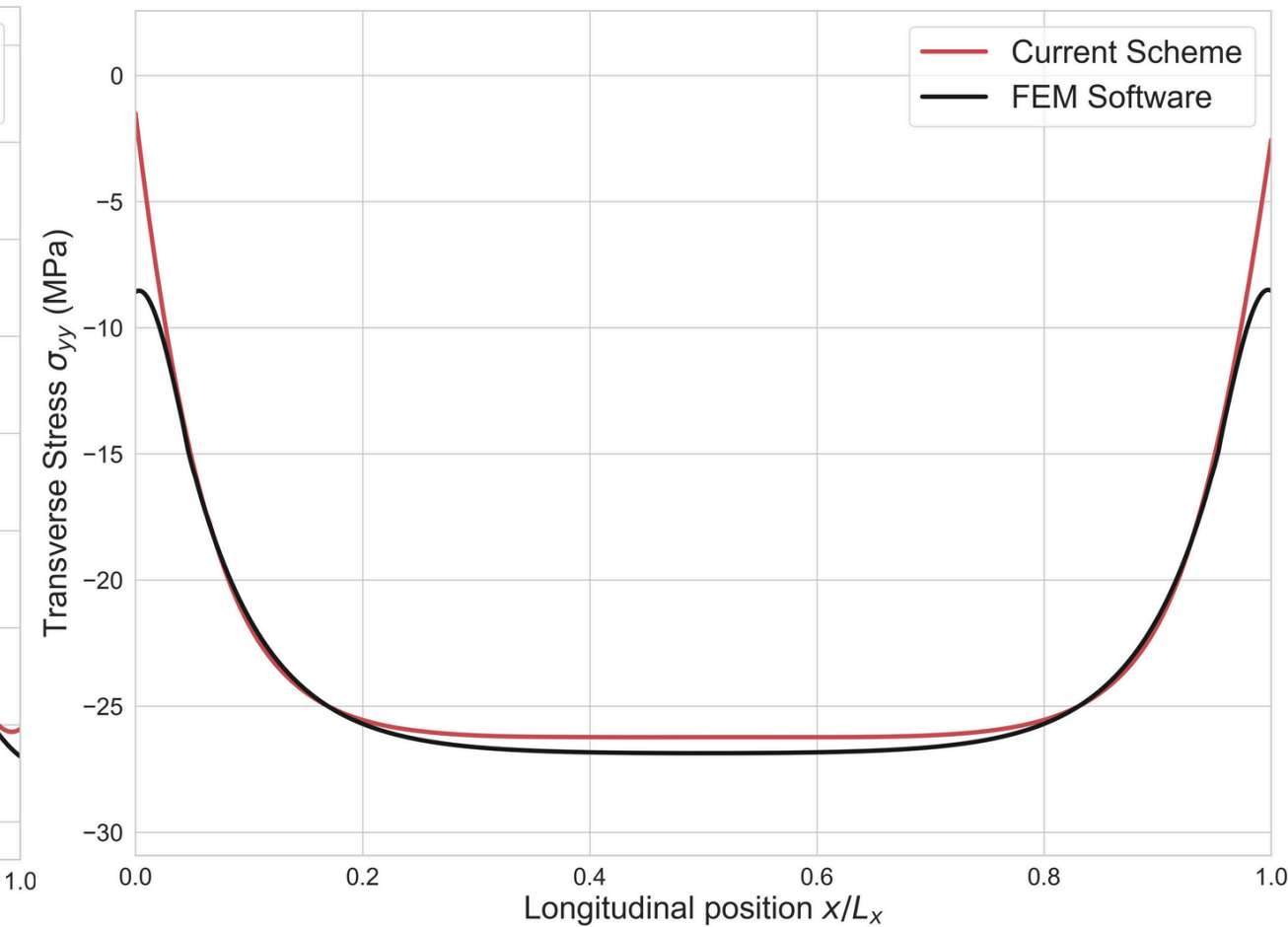
- Axial Stress: **General agreement, small discrepancies near boundaries.**

Comparison of Shear stress at $y/L_y = 0.5$ with FEM



- Shear Stress: **Good alignment, minor variations near boundaries.**

Comparison of Transverse stress at $y/L_y = 0.5$ with FEM



- Transverse Stress: **Close agreement, slight differences in peak values.**

Implementation of the Model

I. The Effect of Temperature-Dependent Properties

Investigates variations in voltage, temperature, and stress distributions when using constant properties, temperature-dependent electrical resistivity, and both temperature-dependent electrical and thermal conductivity.

II. The Effect of Bonding Layer

Compares the standard laminate model to a structure with bonding layers (constant properties) inserted between each layer.

III. The Effect of Effect of Different Bonding Layers

Extends the comparison by evaluating the impact of three different bonding materials on the overall performance.

IV. The Effect of Maximum V_f in Hybrid Laminates

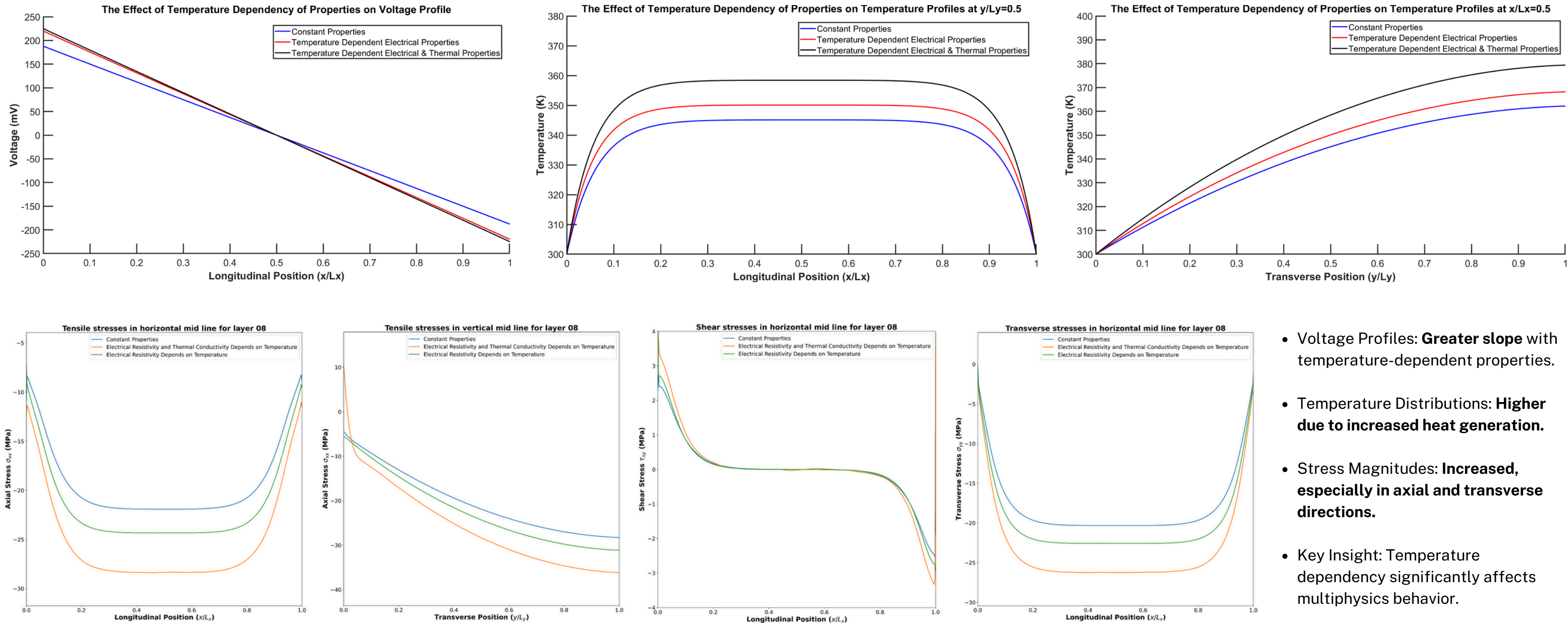
Studies the impact of different maximum filler volume fractions (V_{f_max}) on electrical, thermal, and mechanical responses.

V. The Effect of Gradation ($V_f = m \cdot x^n$)

Examines how varying the exponent ($n = 0.5, 1, 2$) affects filler distribution and composite behavior.

I. The Effect of Temperature-Dependent Properties

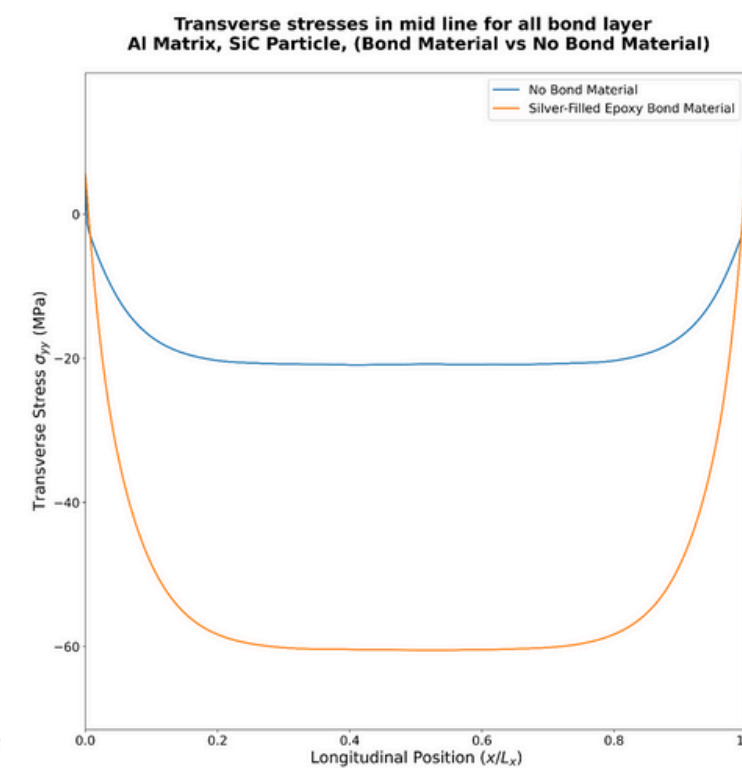
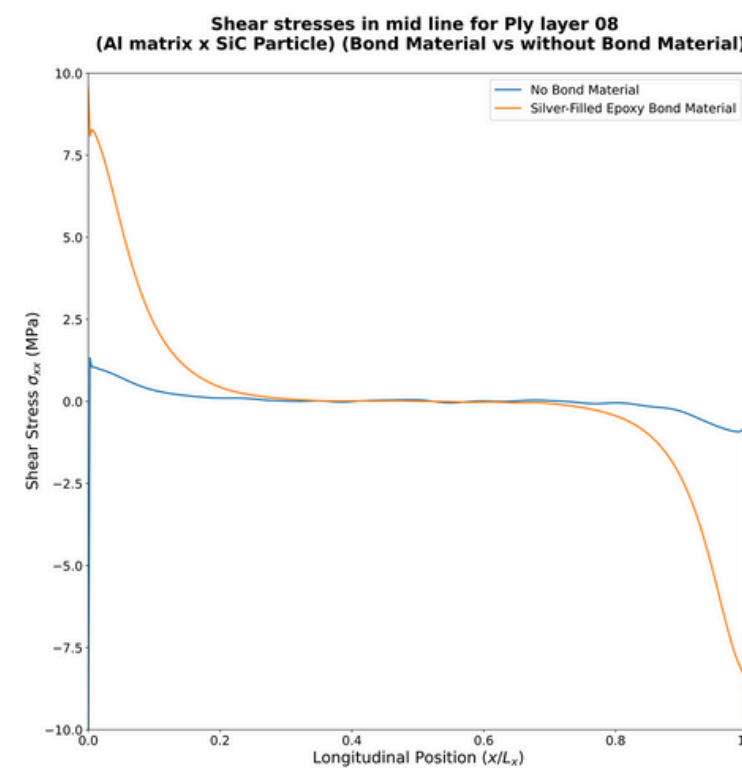
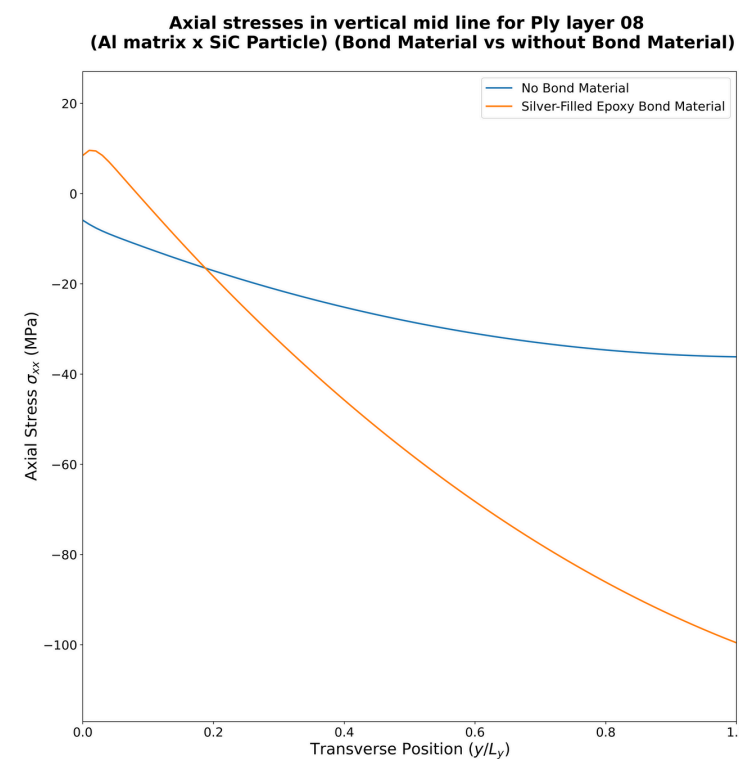
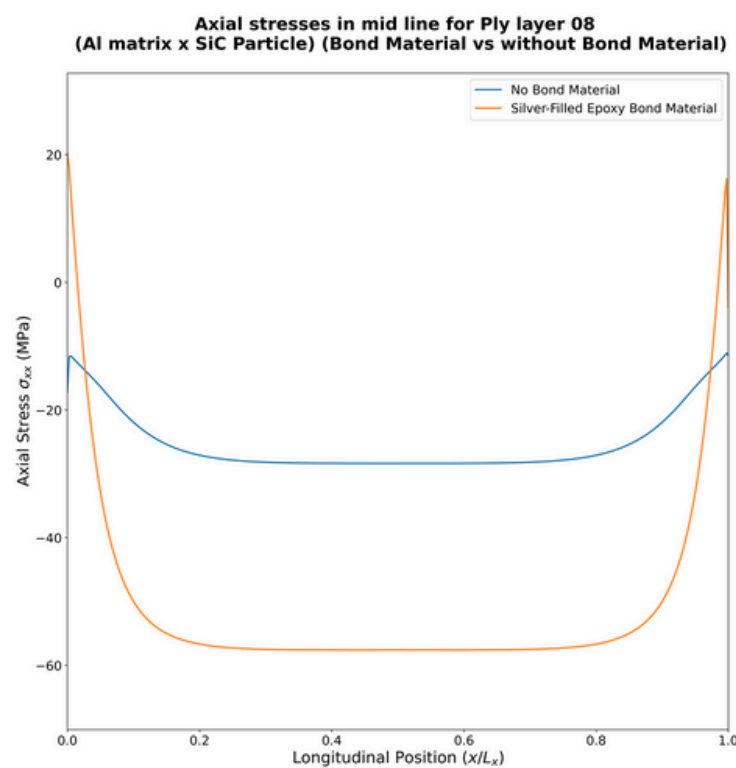
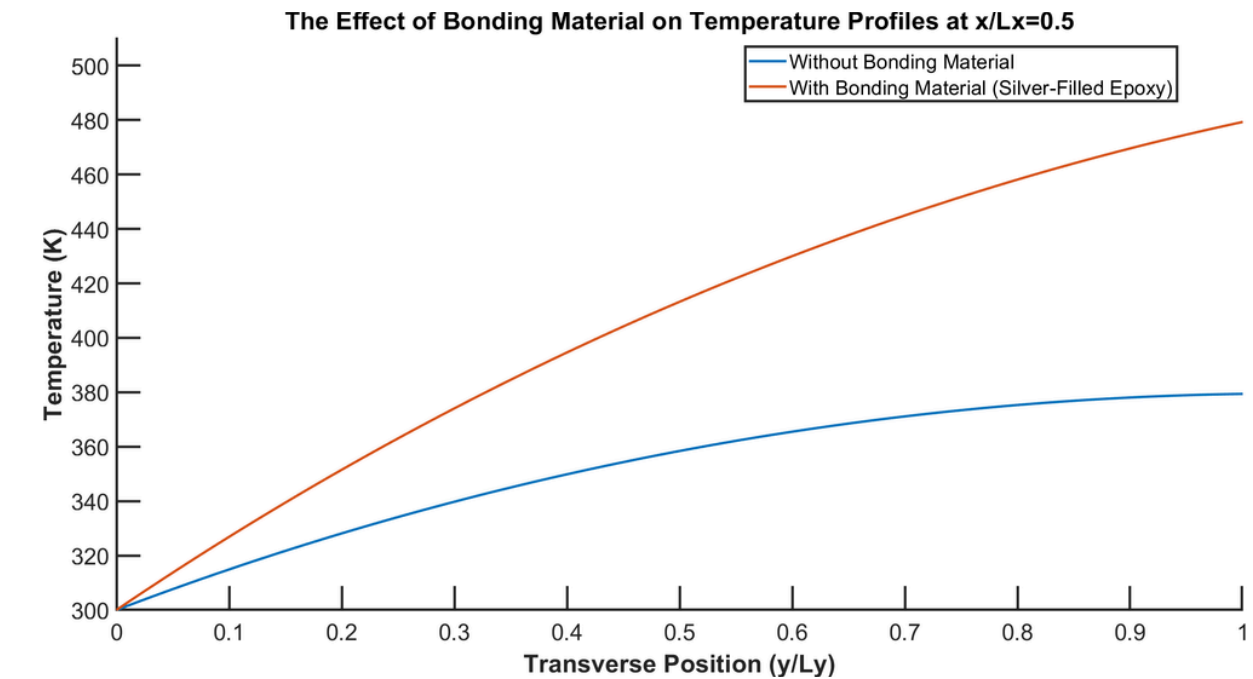
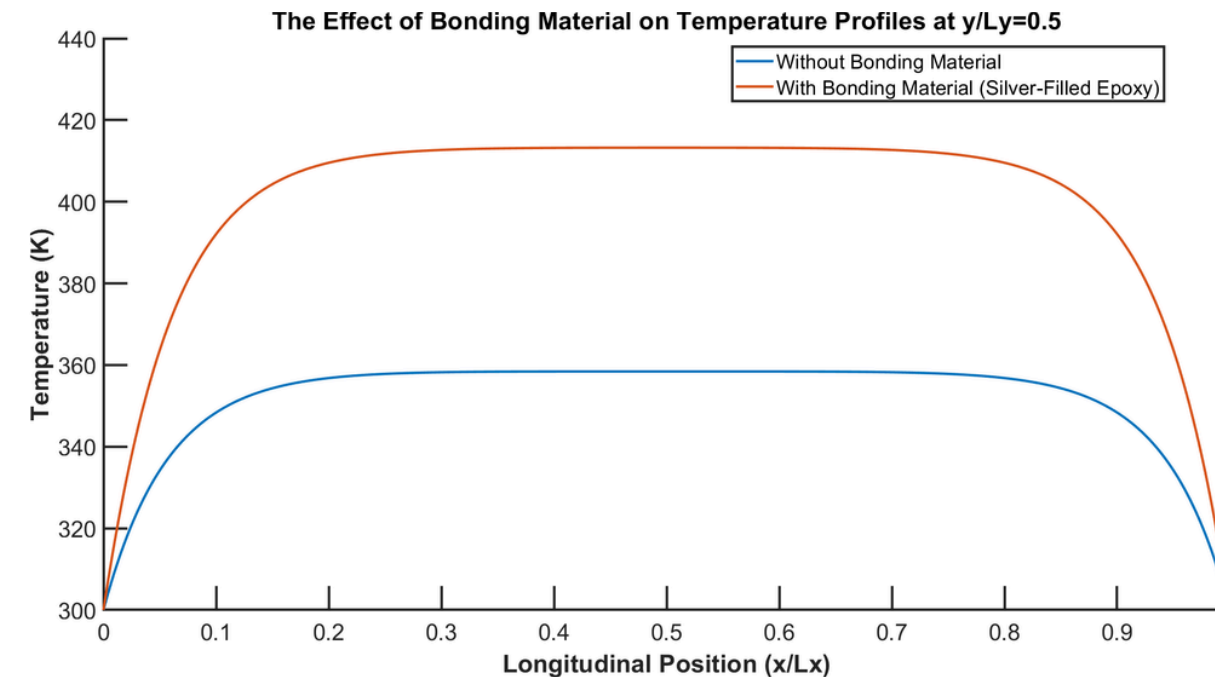
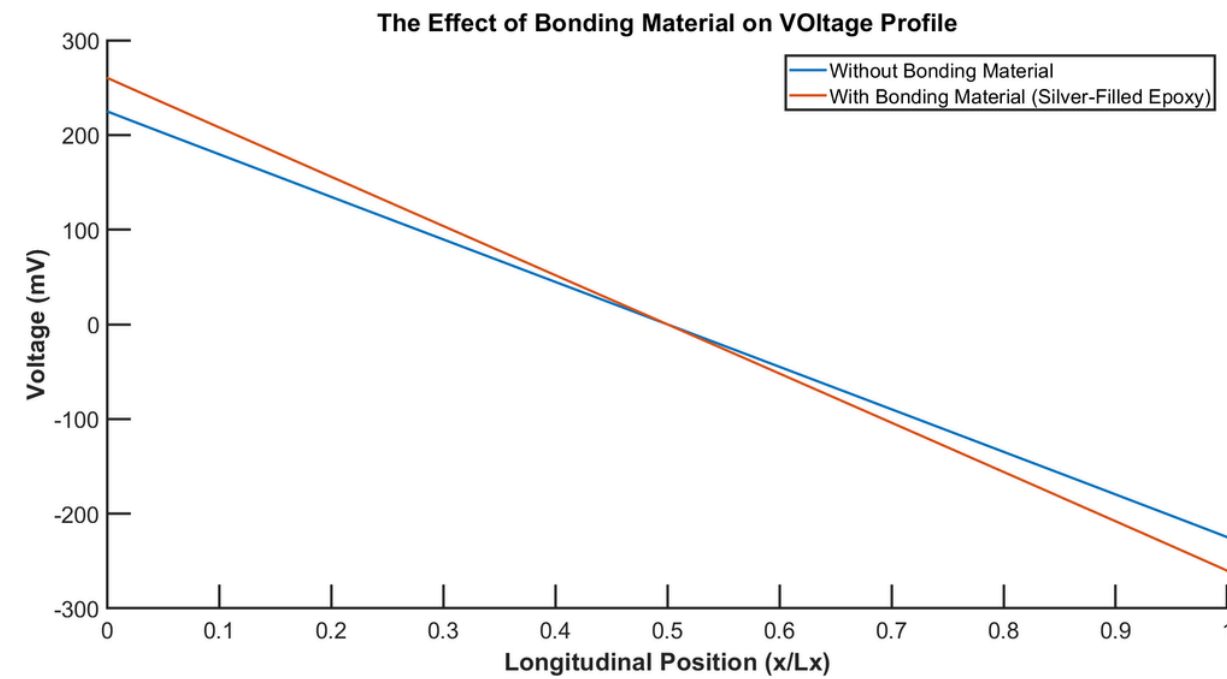
[Al-SiC, $n = 0.5$, $V_f_{\max} = 0.4$]



- Voltage Profiles: **Greater slope** with temperature-dependent properties.
- Temperature Distributions: **Higher** due to increased heat generation.
- Stress Magnitudes: **Increased, especially in axial and transverse directions.**
- Key Insight: Temperature dependency significantly affects multiphysics behavior.

II. The Effect of Bonding Layer

[Al-SiC, $n = 0.5$, $V_f_{\max} = 0.4$]

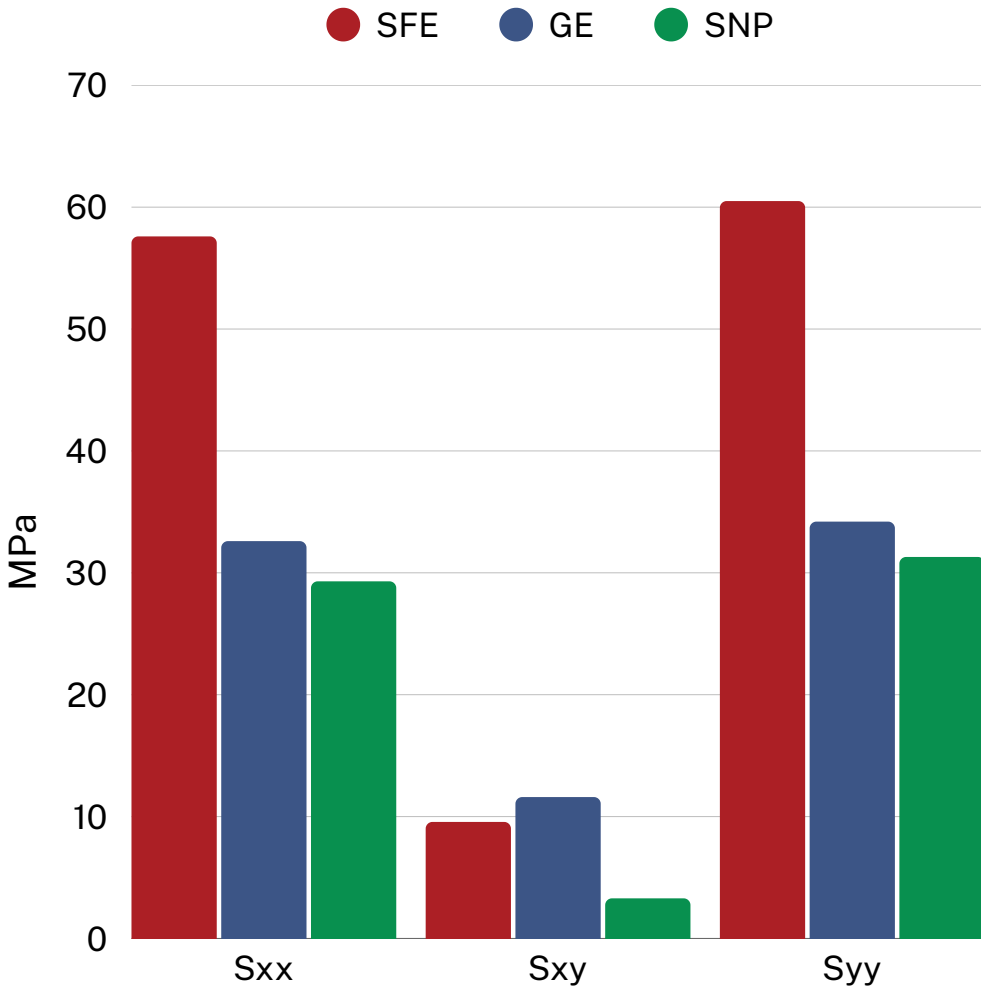
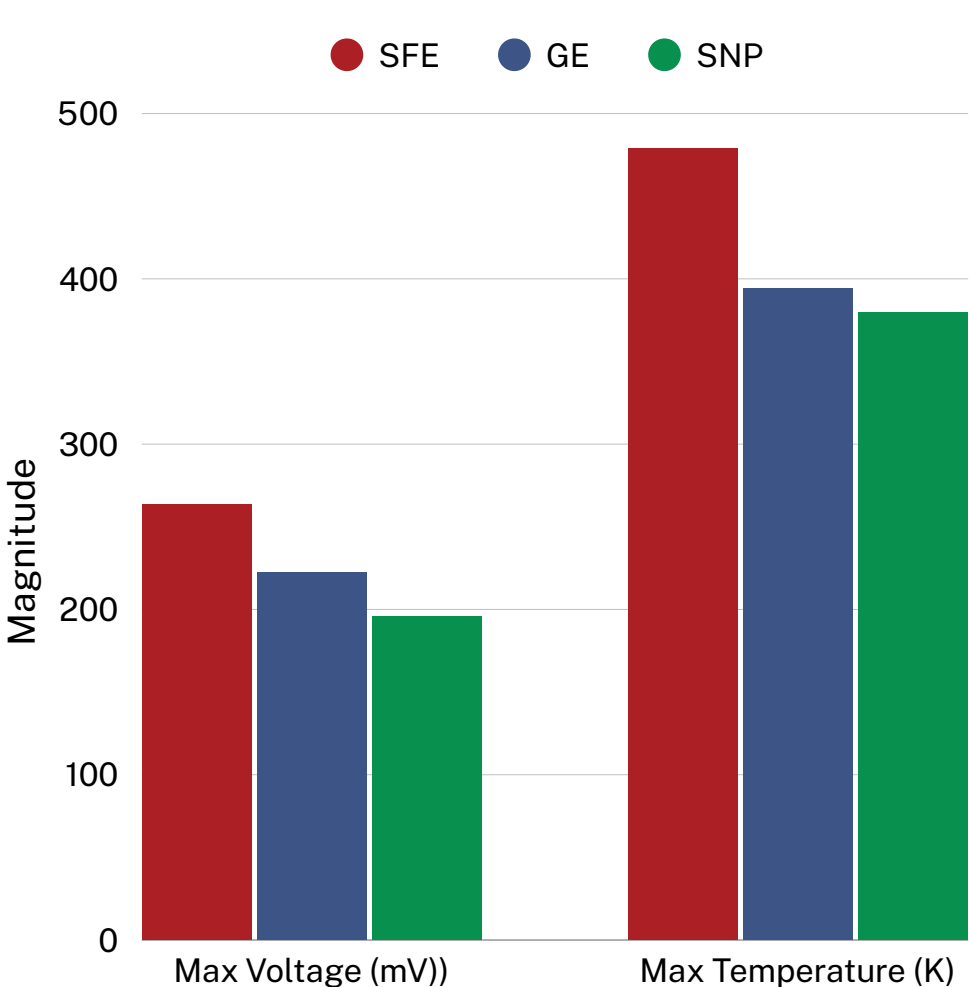


- Voltage Profiles: **Greater slope** with **higher maximum voltage** with bonding material.
- Temperature Distributions: **Higher temperatures** with bonding due to **thermal resistance**.
- Stresses: Can be **very high** with **bonding layer**, affecting mechanical integrity.

III. The Effect of Different Bonding Layers

[Al-SiC, $n = 0.5$, $V_{f_max} = 0.4$]

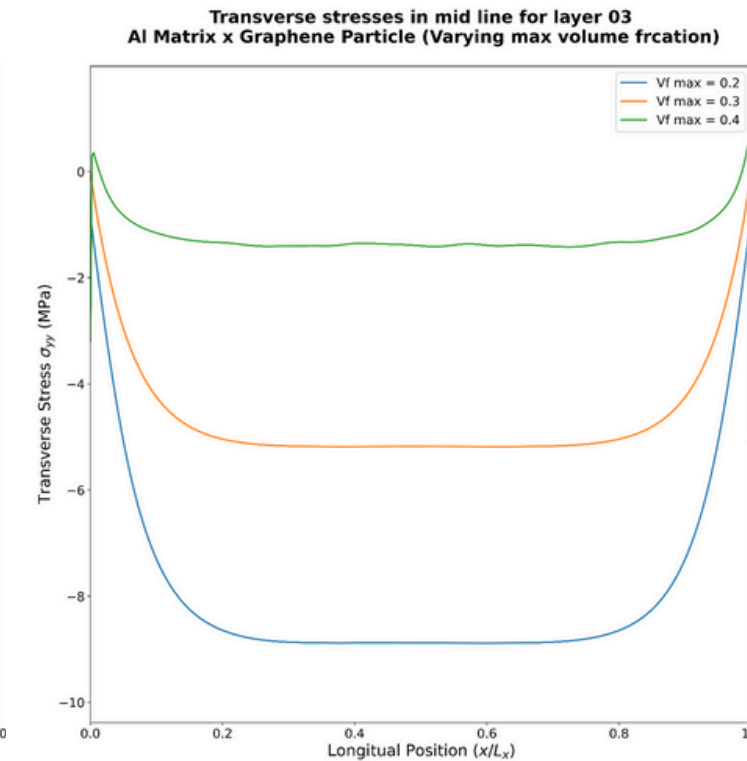
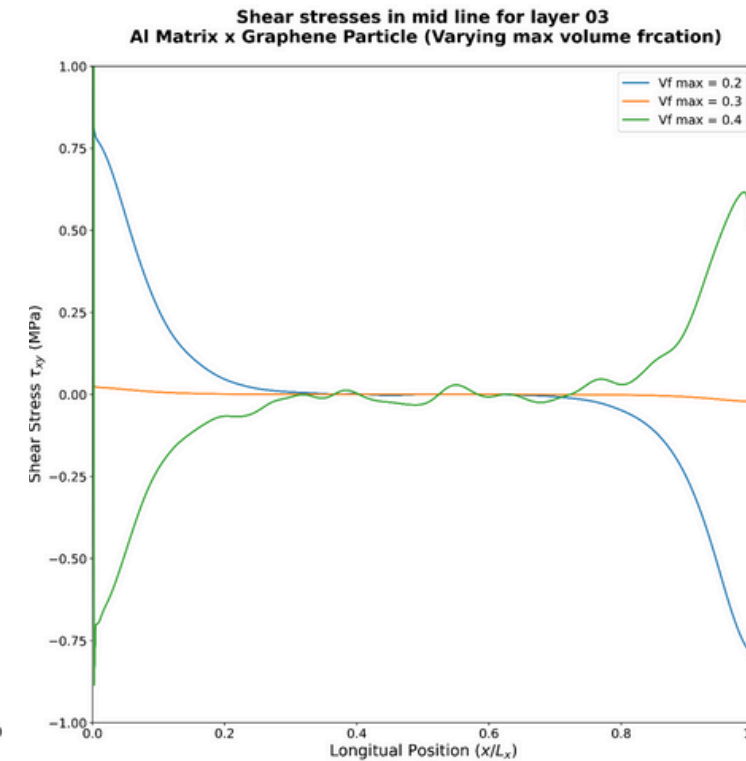
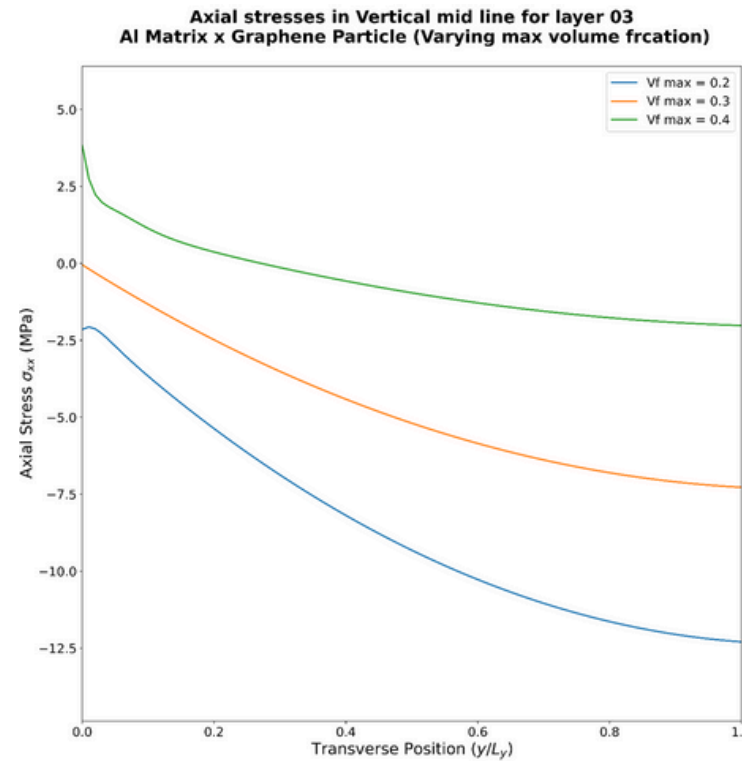
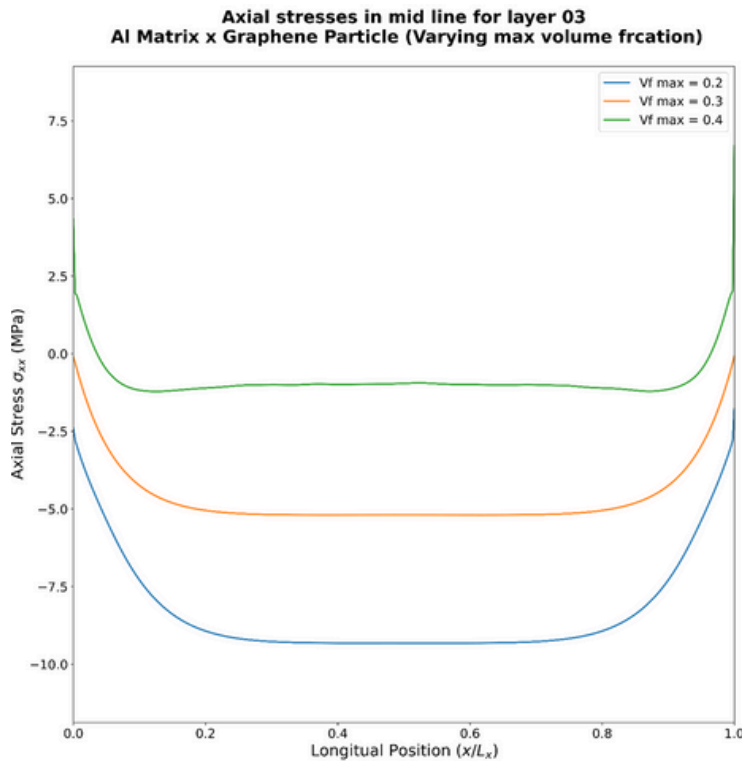
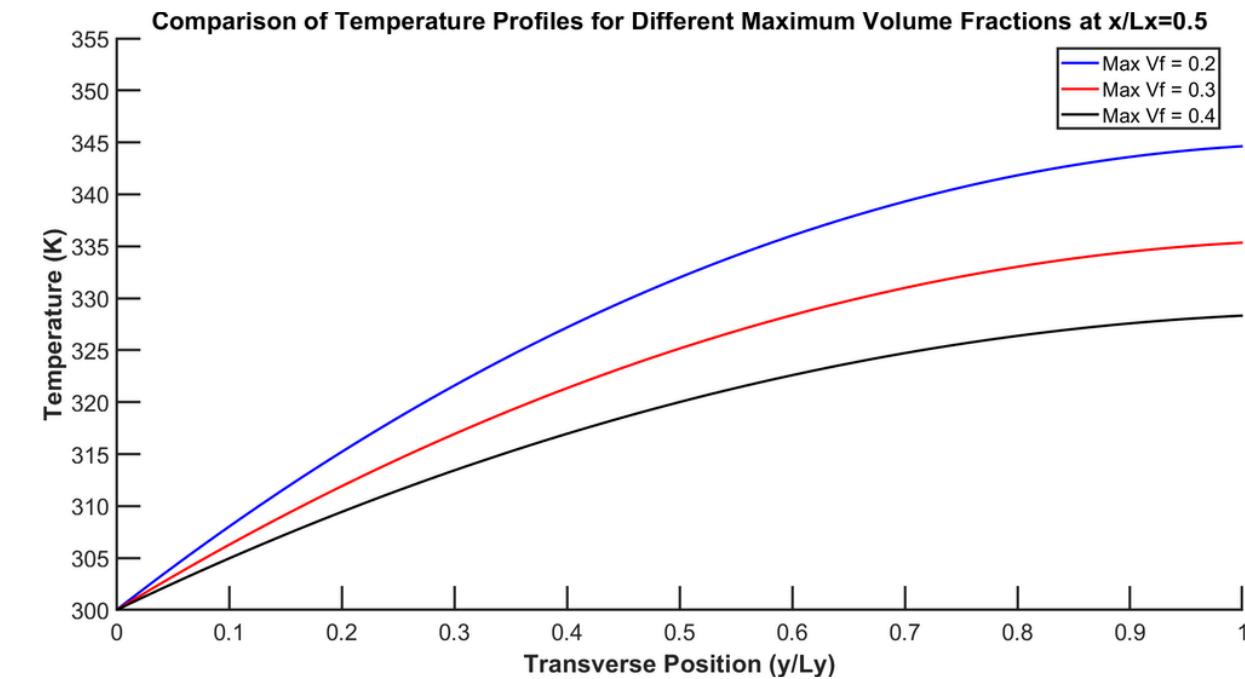
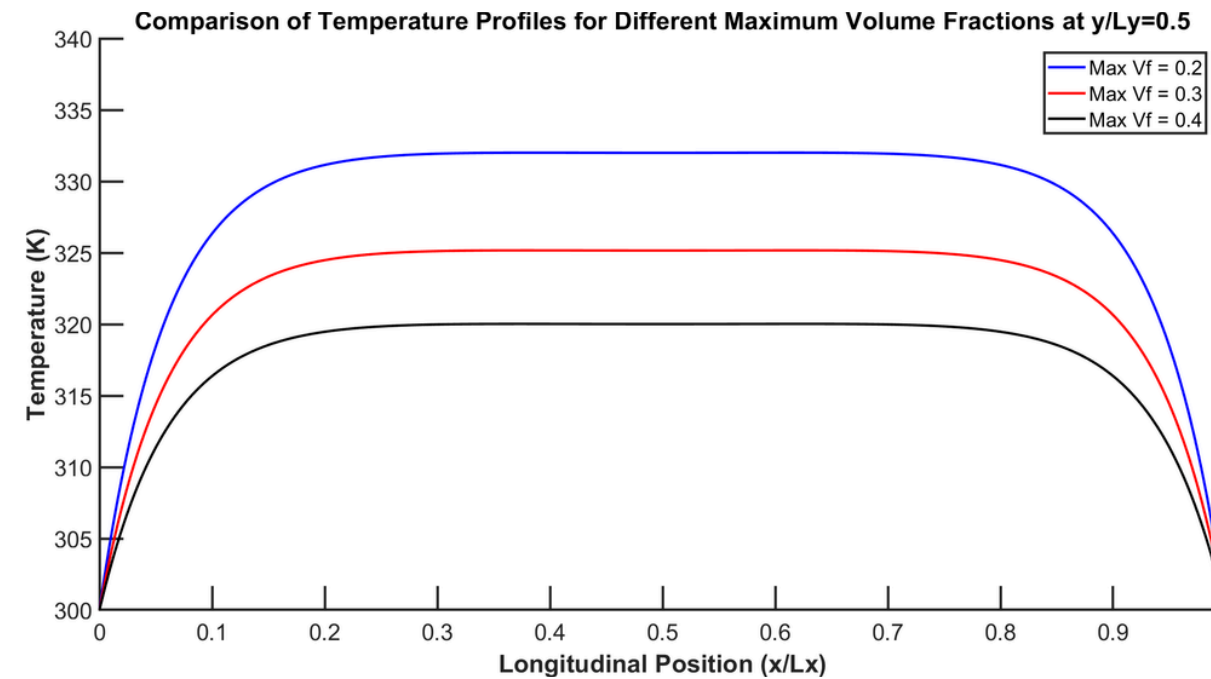
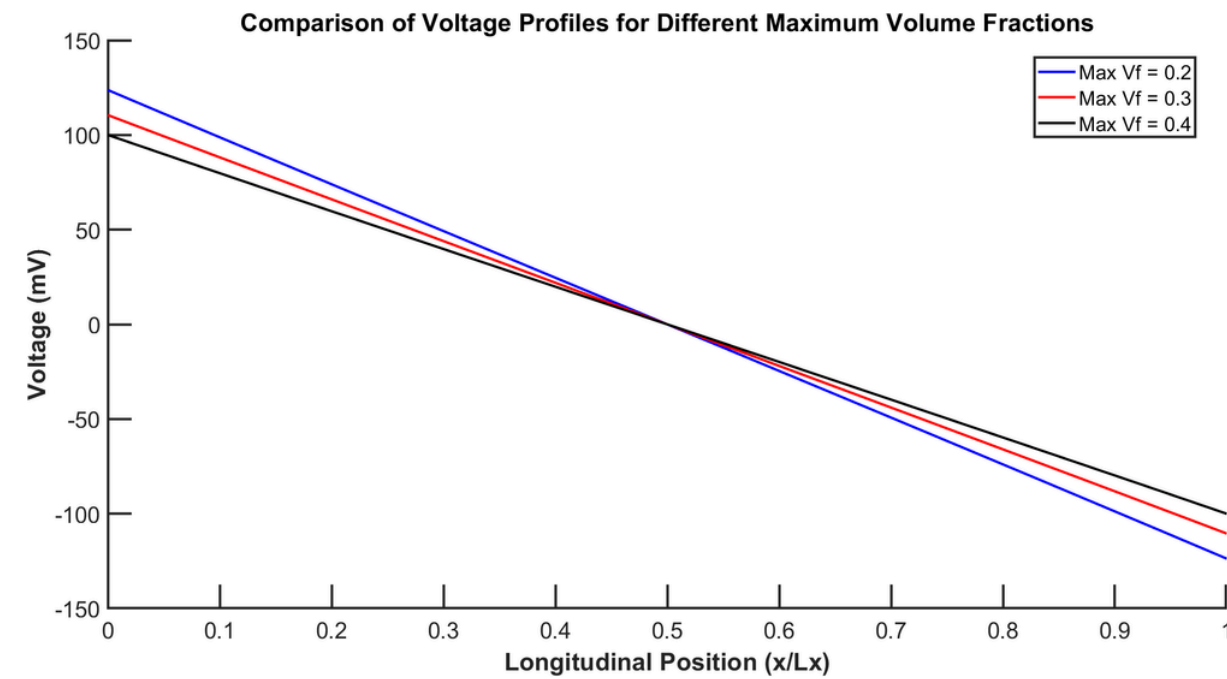
Bond Material	Maximum Voltage (mV)	Maximum Temperature (K)	Maximum σ_{xx} (MPa)	Maximum σ_{xy} (MPa)	Maximum σ_{yy} (MPa)
Silver Filled Epoxy	263.38	479.32	57.6	9.56	60.5
Graphene-Epoxy	222.33	393.93	32.6	11.6	34.2
Silver Nanowire Paste	195.45	379.66	29.3	3.29	31.3



- Voltage Profiles: SNP operates at lower voltages, indicating efficient energy use.
- Temperature Distributions: SNP maintains the lowest temperature, suggesting better heat management.
- Stress Magnitudes: SNP exhibits the lowest axial and transverse stresses, highlighting its mechanical stability.
- Key Insight: **SNP is optimal for minimizing temperature and stress, enhancing overall performance.**

IV. The Effect of Maximum Vf in Hybrid Laminates

[Al-Graphene, $n = 0.5$]

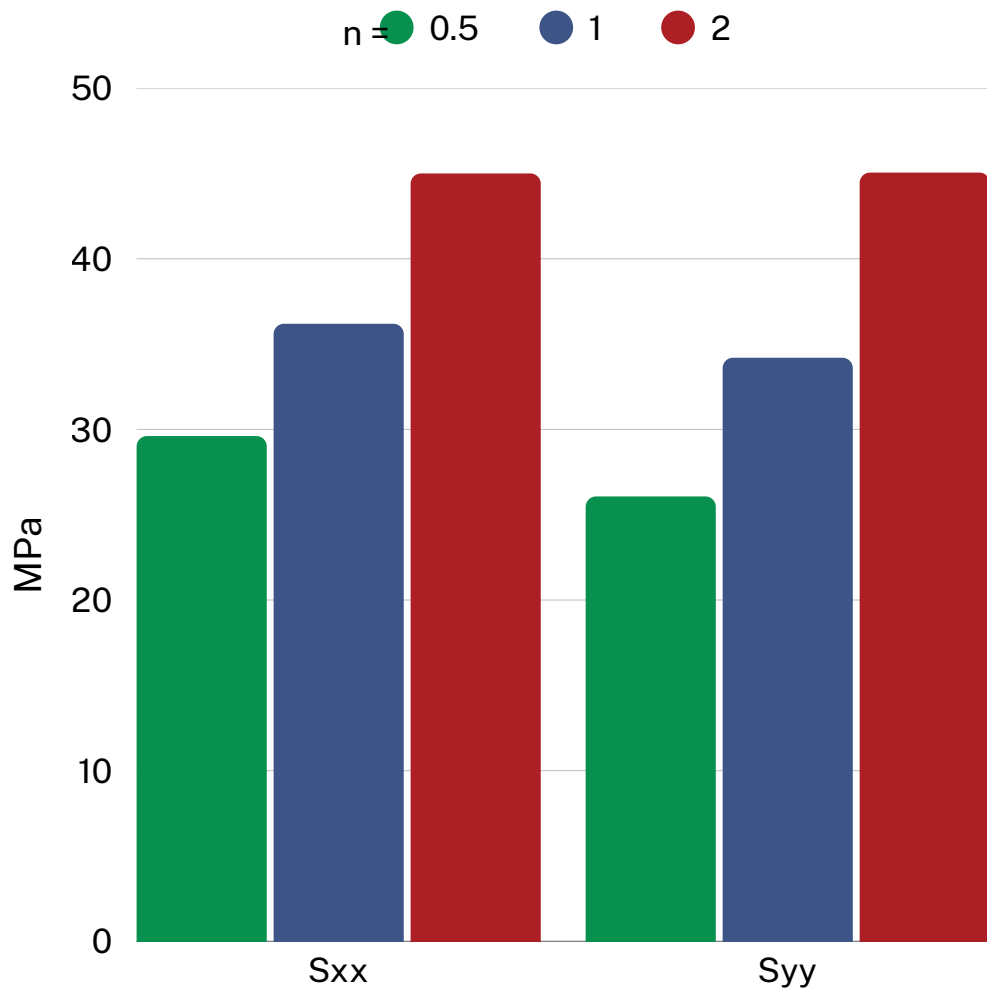
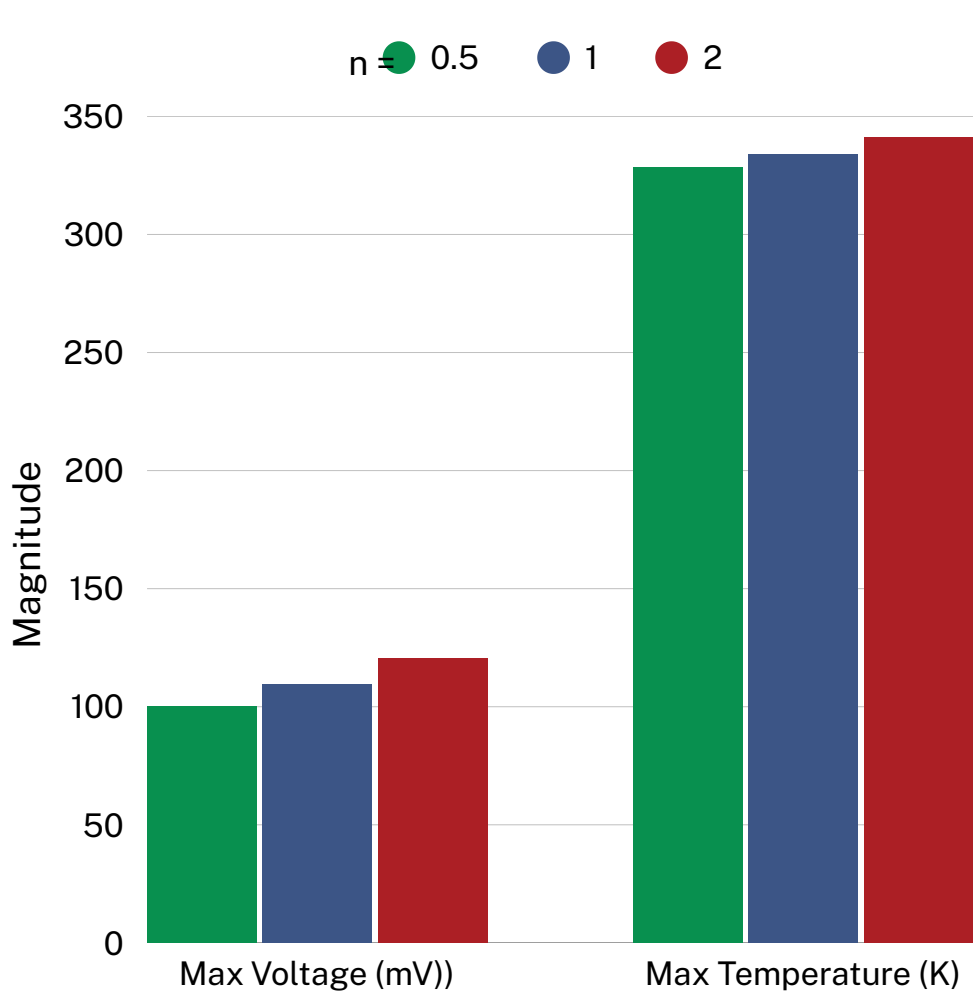


- Voltage Profiles: **Higher Vf reduces voltage drop.**
- Temperature Profiles: **Higher Vf leads to lower temperature rise.**
- Axial Stresses: **Higher Vf increases compressive axial stress.**
- Shear Stresses: **Higher Vf reduces shear stress variations.**
- Transverse Stresses: **Higher Vf decreases peak transverse stress.**

V. The Effect of Gradation ($V_f = m \cdot x^n$)

[Al-Graphene, $V_{f_max} = 0.4$]

Case	Maximum Voltage (mV)	Maximum Temperature (K)	Maximum σ_{xx} (MPa)	Maximum σ_{yy} (MPa)
$n = 0.5$	100.2	328.34	29.61	26.07
$n = 1$	109.03	333.88	36.19	34.2
$n = 2$	120.32	341.29	45.00	45.05



- Voltage and Temperature: **Higher exponent n increases both maximum voltage and temperature**, indicating greater electrical and thermal buildup.
- **Axial Stress and Transverse Stress: Both increase with n** , showing higher mechanical loading with steeper volume fraction gradients.

Research Outcomes

- **Validated Multiphysics Model**

Developed and verified a semi-analytical model for coupled electrical, thermal, and mechanical analysis of hybrid laminated composite conductors under a DC field.

- **Insights on Temperature-Dependent Properties**

Demonstrated the significant impact of temperature-dependent electrical resistivity and thermal conductivity on voltage distribution, heat generation, and stress distribution.

- **Influence of Bonding Layers**

Showed that adding bonding layers affects electrical and thermal performance, with variations depending on material selection.

- **Effect of Filler Distribution (V_f Profile)**

Identified optimal V_f distribution strategies by analyzing different exponents ($n = 0.5, 1, 2$) in the $V_f = m \cdot x^n$ relation.

- **Design Optimization Considerations**

Provided guidelines for selecting filler content, bonding layers, and temperature-dependent properties to enhance performance and reliability.

Conclusion

- **Developed a Novel Semi-Analytical Multiphysics Model** for hybrid laminated composite conductors under a DC field.
- **Integrated Electrical, Thermal, and Mechanical Domains** to capture coupled effects.
- **Implemented Temperature-Dependent Material Properties** for realistic predictions of the behavior.
- **Investigated Key Design Parameters**, including bonding layers, variable volume fractions, and their impact on performance.
- **This thesis provides a new Modeling Framework** for analyzing laminated composites under a DC field efficiently.

Future Prospect

- **Extending to More Complex Composites**

Adapting the model to continuous fiber composites and woven laminates for broader applicability.

- **Transient Analysis**

Expanding the model to transient simulations to capture dynamic effects like thermal cycling and pulsed electrical loading.

- **Investigation of Thermal and Mechanical Failure Modes**

Integrating progressive failure models to predict material degradation, delamination, and ultimate failure under multiphysics loading.

- **Multi-Scale Modeling Approach**

Developing a hierarchical modeling strategy to bridge microscale particle interactions with macroscale composite behavior.

- **Industry Applications & Optimization**

Refining the model for real-world applications in electronics, aerospace, and automotive sectors, optimizing design for performance and reliability.

ME400
PROJECT AND THESIS

THANK YOU

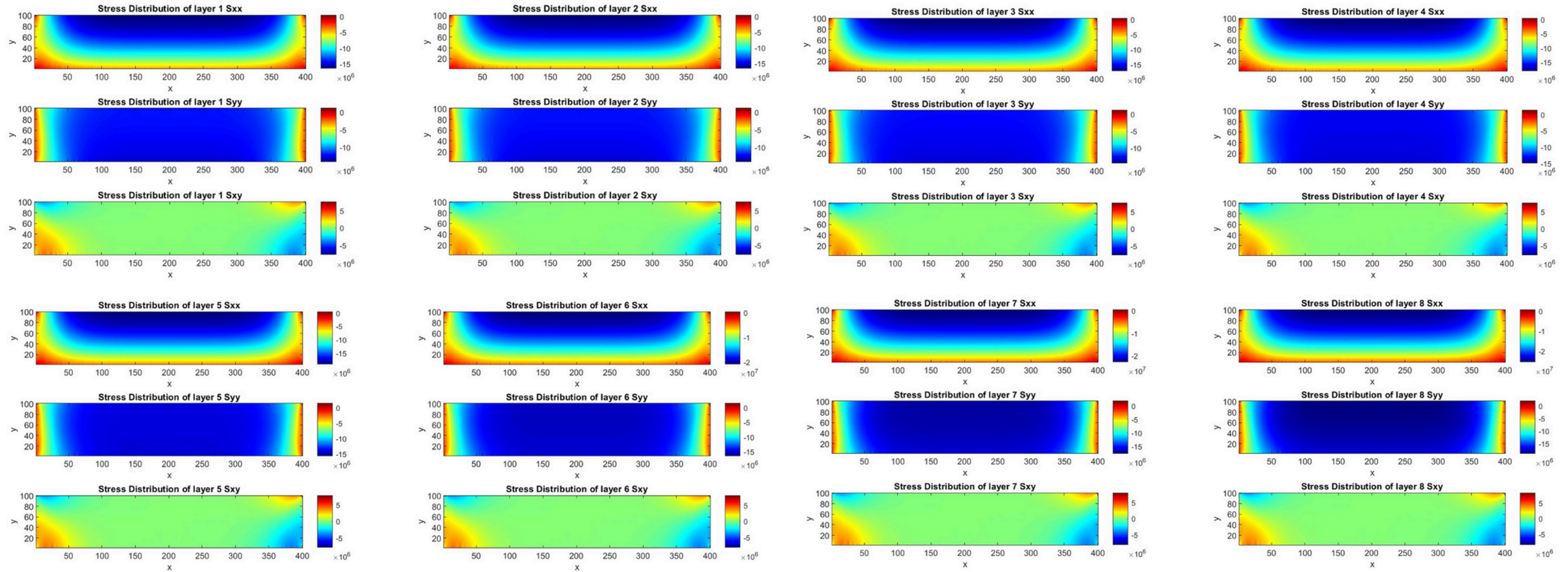


Department of Mechanical Engineering, BUET

Applications of Composite Conductors

- **Power Transmission** – Lightweight, high-conductivity alternatives to conventional metal conductors.
- **Aerospace & Automotive** – Used in lightweight wiring for electric vehicles and aircraft.
- **EMI Shielding** – Protects electronics from signal interference.
- **Wearable & Flexible Electronics** – Enables durable, stretchable circuits.
- **Energy Storage** – Enhances supercapacitor and battery performance.
- **Smart Grids & Renewables** – Improves efficiency in solar and wind power systems.
- **High-Temperature Applications** – Stable in extreme thermal environments like reactors and space.

Layer-wise Stress Distribution



Material Properties

Material	Electrical Resistivity ($\Omega \cdot \text{m}$)	Thermal Conductance ($\text{W/m} \cdot \text{K}$)	Temp. Coeff. of Resistivity	Modulus of Elasticity (GPa)	Poisson's Ratio
Aluminum	2.82×10^{-8}	232.1	4.29×10^{-3}	70.0	0.33
Graphene	1×10^{-8}	2732.3	2.20×10^{-3}	1000.0	0.165
Silicon Carbide (SiC)	1.00×10^3	844	-1.00×10^{-3}	410.0	0.17-0.19
Silver-Filled Epoxy (SFE)	1.00×10^{-6}	10	-	5-10	0.3-0.35
Graphene-Epoxy Composites (GEC)	1.00×10^{-7}	50	-	0.1-1	0.35-0.4
Silver Nanowire Pastes (SNP)	5.00×10^{-8}	100	-	4-6	0.3-0.4

